Explainable AI-Based Comparative Analysis of Deep Learning and Large Language Models for Symptom Based Disease Prediction

MAJOR PROJECT REPORT submitted in partial fulfillment of the requirements for the Award of the Degree of

BACHELOR OF TECHNOLOGY

In

INFORMATION TECHNOLOGY

By

Aravind Gurram 228W1A1290

Srinivasa Bhargav Pinjala 228W1A12B7

Under the Guidance of

Dr.C.S.Pavan Kumar

Sr. Assistant Professor

DEPARTMENT OF INFORMATION TECHNOLOGY

V R SIDDHARTHA ENGINEERING COLLEGE

(AUTONOMOUS - AFFILIATED TO JNTU-K, KAKINADA)

Approved by AICTE &Accredited by NBA

KANURU, VIJAYAWADA-520007

ACADEMIC YEAR

(2025-26)

V.R.SIDDHARTHA ENGINEERING COLLEGE

(Affiliated to JNTUK: Kakinada, Approved by AICTE, Autonomous)
(An ISO certified and NBA accredited institution)
Kanuru, Vijayawada – 520007

CERTIFICATE

This is to certify that this project report titled **"Explainable AI-Based Comparative Analysis of Deep Learning and Large Language Models for Symptom-Based Disease Prediction"** is a Bonafide record of work done by **ARAVIND GURRAM (228W1A1290), SRINIVASA BHARGAV PINJALA (228W1A12B7)** under my guidance and supervision is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Information Technology, **V.R. Siddhartha Engineering College** (Autonomous under JNTUK) during the year 2025-26.

(Dr. C. S. Pavan Kumar)
Sr. Assistant Professor
Dept. of Information Technology

(Dr. M. Suneetha)
Professor &Head
Dept. of Information Technology

EXTERNAL EXAMINER SIGNATURE

DEPARTMENT OF INFORMATION TECHNOLOGY

V.R. SIDDHARTHA ENGINEERING COLLEGE

PROJECT SUMMARY

S.No	Item	Description
1	Project Title	Explainable AI-Based Comparative Analysis of Deep Learning and Large Language Models for Symptom-Based Disease Prediction
2	Student Names & Numbers	Aravind Gurram(228W1A1290) Srinivasa Bhargav Pinjala(228W1A12B7)
3	Name of The Guide	Dr. C. S. Pavan Kumar
4	Research Group	Artificial Intelligence and Machine Learning
5	Application Area	Health care
6	Aim of the Project	To build a system that can predict diseases based on user symptoms using Deep Learning and LLM models, and also explain the results clearly using SHAP.
7	Project Outcomes	1. Predicts possible diseases based on the symptoms given by the user. 2. Provides results with good accuracy using trained models. 3. Gives clear explanations of predictions using SHAP for better understanding.

Student Signatures

1.
2.

Signature of the Guide

ACKNOWLEDGEMENT

First and foremost, I sincerely salute our esteemed institution **V.R SIDDHARTHA ENGINEERING COLLEGE** for giving me this opportunity for fulfilling my project. I am grateful to our Principal **Dr. A.V.RATNA PRASAD**, for his encouragement and support all through the way of my project.

On the submission of this Project report, I would like to extend my honour to **Dr. M.Suneetha**, Head of the Department, IT for her constant motivation and support during the course of my work.

I feel glad to express my deep sense of gratefulness to my project guide **Dr.C.S.Pavan Kumar, Sr.Assistant Professor** for **his** guidance and assistance in completing this project successfully.

I would also like to convey my sincere indebtedness to all faculty members, including supporting staff of the Department, friends and family members who bestowed their great effort and guidance at appropriate times without which it would have been very difficulty on my part to finish the project work.

TABLE OF CONTENTS

ACKNOWLEDGEMENT	iv
LIST OF FIGURES	vi
LIST OF TABLES	vii
LIST OF SYMBOLS	viii
LIST OF EQUATIONS	ix
ABSTRACT	x
CHAPTER-1 Introduction	1-6
1.1Origion of the Problem	1
1.2 Basic definitions and Background	2
1.3Problem Statement with Objectives and Outcomes	5
1.4 Related Applications	6
CHAPTER-2 Review of Literature	7-12
2.1Description of Existing Systems	7
2.3Summary of Literature Study	9
CHAPTER-3 Proposed Method	13-18
3.1Design Methodology	13
3.2System Architecture Diagram	16
3.3Description of Algorithms	19
3.4 Description of datasets, Requirements and Tools	24
CHAPTER-4 Results and Observations	26-55
4.1Stepwise description of Results	26
4.2Test case results/Result Analysis	48
4.3Observations from the work	53
CHAPTER-5 Conclusion and Future work	56-60
5.1 Conclusion	56
5.2 Future study	59
References	61
Appendix	63

LIST OF FIGURES

Figure		Page No
Fig 3.2.1	Architecture Diagram for DL Models	16
Fig 3.2.2	Architecture Diagram for LLM Models	17
Fig 4.1.1	Graphs for ELECTRA Model	27
Fig 4.1.2	Graphs for ERNIE Model	29
Fig 4.1.3	Graphs for XLM-RoBERTa Model	31
Fig 4.1.4	Graphs for XLNet + ELECTRA Model	34
Fig 4.1.5	Graphs for XLM-RoBERTa + ERNIE Model	36
Fig 4.1.6	Graphs for ELECTRA + ERNIE Model	39
Fig 4.1.7	Graphs of Deep Learning Models	43
Fig 4.1.8	Graphs of Large Language Models	45
Fig 4.1.9	XAI (SHAP) Analysis for Deep Learning Model	47
Fig4.1.10	XAI (SHAP) Analysis for LLM Model	48
Fig 4.2	Result Analysis for all Models of DL and LLM	53
Fig 4.3.1	Comparison of Results for all Models	54
Fig 4.3.2	Grouped Comparison of Model Categories	55
Fig 6	Code Snippet of LLM model	64

LIST OF TABLES

Table Name		Page No
Table 2.2.1	Literature Review	08
Table 4.1.1.1	Sample Classification Report for ELECTRA Model	27
Table 4.1.1.2	Classification Metrics for ELECTRA Model	27
Table 4.1.2.1	Sample Classification Report for ERNIE Model	29
Table 4.1.2.2	Classification Metrics for ERNIE Model	30
Table 4.1.3.1	Sample Classification Report for XLM-RoBERTa Model	31
Table 4.1.3.2	Classification Metrics for XLM-RoBERTa Model	34
Table 4.1.4.1	Sample Classification Report for XLNet + ELECTRA Model	34
Table 4.1.4.2	Classification Metrics for XLNet + ELECTRA Model	35
Table 4.1.5.1	Sample Classification Report for XLM-RoBERTa + ERNIE Model	36
Table 4.1.5.2	Classification Metrics for XLM-RoBERTa + ERNIE Model	37
Table 4.1.6.1	Sample Classification Report for ELECTRA + ERNIE Model	39
Table 4.1.6.2	Classification Metrics for ELECTRA + ERNIE Model	40
Table 4.2.1	Result Analysis for LLM Models (Mini Project 1)	49
Table 4.2.2	Result Analysis for Hybrid Models (Mini Project 2)	50
Table 4.2.3	Performance Comparison of DL and LLM Models	51

LIST OF SYMBOLS

Symbol	Symbol Name
TP	True Positives
FP	False Positives
FN	False Negatives
TN	True Negatives
Precision (P)	Accuracy of positive predictions
Recall (R)	Ability to find all positive cases
F1-Score	Harmonic mean of Precision and Recall
Accuracy	Overall correctness of model
SHAP	SHapley Additive Explanations
DL	Deep Learning
LLM	Large Language Model
MLP	Multilayer Perceptron

LIST OF EQUATIONS

Equation	Brief Description	Page No
Eq (1)	Precision Formula	26
Eq (2)	Accuracy Formula	26
Eq (3)	Recall Formula	27
Eq (4)	F1-Score Formula	27

ABSTRACT

It is a disease Prediction based on symptoms, which helps in providing suitable precautions and identifying appropriate medications at an early stage. With a rapid growth in advanced AI such as the Deep Learning and Large Language Models providing the strong capability to classify the tasks. However, selecting the suitable model for the structure symptoms data to ensure interpret and still remains a significant challenge.

This project shows a comparative analysis between the dee learning and large language models for a symptom-based disease detection and which can help to prevent the disease in the early stage. The study used the three dl models such as multilayer perception (MLP), TabNet and Resid- ual MLP which are effective for structured table data. For the Large Lan- guage models such as the XLM Roberta, ERNIE and ELECTRA which capture the relationships through the textual representations.

To enhance model transparency, we used explainable artificial-intelligence (XAI) technique using the Shapley Additive Explanations (Shap) model is applied to all the models identify and contribution of individual symptoms in disease prediction. These models are generally evaluated using the standard performance metrics such as accuracy, precision, recall and f1-score. Based on the comparative analysis on among both DL models and LLM models across all models to identify the overall better perform- ing approach. And the results of these study provide insights with the strength and limitations of both model categories to development of AI -driven medical decision support system.

Keywords: Deep Learning, Large language models, XLM Roberta, Shap, Explainable Artificial Intelligence, Disease, TabNet, MLP, Health Analysis.

Chapter – 1

Introduction

1.1 Origin of the problem

The rise in population and diseases has resulted in various problems within the healthcare industry. Disease prediction plays an important role in minimizing mortality rates. However, in some instances, patients fail to consult doctors since they are unaware of the signs, lack access to hospitals, or find it inconvenient. As a result, the diseases become more severe with delayed diagnosis.

Through technological innovations, particularly the development of Artificial Intelligence(AI), there is chance to create automated systems that would help predict diseases based on the presented symptoms. Currently, doctors depend on manual diagnosis, which is usually lengthy and has room for errors. Furthermore, most algorithms developed from machine learning and deep learning operate like black boxes without explanations of their workings. It makes them unreliable.

Over the years, LLMs have been proven efficient in processing information and detecting patterns. Nonetheless, the efficiency of LLMs in processing structured symptom data compared to deep learning is yet to be determined. Another issue concerns the implementation of AI in healthcare, which should be transparent and interpretable.

Beyond the DL approaches, transformer- based LLMs have gained wide range attention for their ability in capturing contextual and semantic connections with in the data. LLMs such as the XLM- Roberta, ERNIE and ELECTRA have been successfully applied in various classifications and also used in the NLP tasks. By converting the symptom data into the textual representations, these models can leverage contextual understanding to improve the prediction performance. Despite their strengths, LLMs are intensive and require careful evaluation when they are applied on the structured dataset.

A major limitation of AI-based healthcare systems has the lack of interpretability. Most of the models are operate on the black-box systems, providing the predictions without explaining the any reason behind them. In medical applications, there is a lack of transparency that can reduce trust. In this case, Explainable Artificial Intelligence (XAI) techniques will help to solve this, such as Shapley Additive Explanations (SHAP), have been introduced to solve by identifying the contribution of individual features, thereby improving the interpretability and reliability of model predictions.

This work main goal is to perform a comparative analysis of Deep Learning models and LLM models for disease prediction. The study evaluates MLP, TabNet, and Residual MLP beside of XLM-RoBERTa, ERNIE, and ELECTRA, while incorporating SHAP for explainability. The comparison study is conducted be- tween multiple levels to identify the most accurate, efficient, and interpretable model [4].

Thus, there is a need for developing a reliable and interpretable disease prediction system. The proposed project aims at designing such an algorithm. Specifically, the paper will involve comparisons of deep learning models and large language models, followed by Explainable AI implementations to ensure credibility and reliability.

1.2 Basic definitions and Background

1.2.1 Disease Prediction Based on Symptoms

The disease prediction based on the symptoms denotes identification of the possible diseases according to the symptoms provided by the patient. In conventional medicine, the prediction is made based on the skills and expertise of the medical professional. On the other hand, AI models are used to automate same process based on symptom data.

1.2.2 Artificial Intelligence (AI)

AI stands for the application of technologies that allow machines to perform cognitive activities which require intelligence. Some of the examples include learning, reasoning, and decision making. In medicine, AI can be applied to perform tasks related to analyzing patient information, helping with diagnosing diseases, etc.

1.2.3 Deep Learning (DL)

The Deep Learning is an area of AI, where neural networks consisting of more than one layer are used for data analysis. The DL models included in this project include Multilayer Perceptron (MLP), TabNet, and Residual MLP.

1.2.4 Large Language Models (LLM)

The Large language models denotes AI models that understand natural languages due to the usage of transformer architectures. Such models include XLM-RoBERTa, ERNIE, and ELECTRA. These models will help to analyze symptom data and make a diagnosis by transforming them to textual representation first.

1.2.5 Transformer Models

The transformer models refer to deep learning architectures which are designed for working with sequential data, particularly text data. These models apply attention mechanisms for recognizing connections within words. All large language models included in this project belong to the transformer family.

1.2.6 Explainable Artificial Intelligence (XAI)

Explainable AI involves methods for making AI models' decisions understandable for people. In this project, SHAP values (Shapley Additive Explanations) are used in interpret model predictions based on symptoms and provide explanations for each of them.

1.2.7 Multilayer Perceptron (MLP)

A Multilayer Perceptron is also one of the deep learning algorithms based on feedforward neural networks consisting of several layers of neurons. The model includes input, hidden, and output layers. It's widely applied for different types of classifications. MLP is applied in this project to train model in predicting the correlation between input symptom and corresponding disease.

1.2.8 TabNet

TabNet is a deep learning model that is specifically designed to work with tabular data. And it uses a sequential attention mechanism that helps to focus on most useful features for training purposes. In this case study, TabNet will be used to effectively work with structured symptom data.

1.2.9 Residual MLP

Residual MLP (Multilayer Perceptron) is a type of MLP architecture with residual connections to improve training results. Residual MLP avoids the problem of vanishing gradients and enables effective learning of deeper neural networks. In this research, Residual MLP will be used to increase prediction accuracy.

1.2.10 XLM-RoBERTa

XLM-RoBERTa is a transformer-based Large Language Model that can work with multilingual text. It learns complex contexts from large multilingual datasets. XLM-RoBERTa will be used to predict diseases based on textual representation of symptom data.

1.2.11 ERNIE

ERNIE (Enhanced Representation through Knowledge Integration) is a transformer-based language that integrates external knowledge in order to increase effectiveness. ERNIE can effectively capture

semantic and contextual relationships. ERNIE will be used in this project to predict diseases by understanding symptom relationships.

1.2.12 ELECTRA

ELECTRA is a transformer-based model that increases the speed and effectiveness of training thanks to replaced token detection instead of traditional masking. ELECTRA is computationally inexpensive and can achieve great results in classification tasks. It will be used for predicting diseases based on symptoms' textual description.

1.3 Problem Statement with Objectives and Outcomes

Artificial Intelligence became widely used in healthcare due to its capabilities for disease prediction and diagnostics. With development of machine learning, models can analyze large medical datasets, recognize patterns between symptoms and diseases, and detect illnesses at an early stage.

Deep Learning models demonstrated impressive results in processing structured medical data, whereas Large Language Models proved to be highly effective in analyzing textual data and recognizing context. Choosing the right model for predicting diseases based on symptoms is still a problem.

One more serious issue connected with application of Artificial Intelligence to healthcare is lack of interpretability of models that act as black boxes. To solve the problem, Explainable AI techniques such as SHAP can be used.

Thus, this research will compare several Deep Learning and Large Language Models for disease prediction that integrates Explainable AI technology to increase model interpretability.

1.4 Related Applications

There are a lot of AI applications nowadays which can provide insight regarding health problems of an individual. It could be described as symptom checker apps, where the application provides users with possible diseases based on input symptoms. Some examples of such applications include WebMD, Ada Health etc. However, the majority of such algorithms rely on predetermined logic so their output might sometimes be inaccurate or non-personalized.

Chatbot-based applications provide a conversational interface, where the user talks with the application which tries to understand what is happening and provide corresponding outputs. Chatbot algorithms are useful but fail to perform properly when the input information provided by users is quite complicated or vague.

Apart from that, numerous applications employ ML/Deep Learning-based models which provide significantly more accurate predictions. However, this approach raises the problem of model interpretability since there is no transparent way of explaining the decisions of such applications.
Our goal in this research is to enhance mentioned aspects and create a hybrid model, based on Deep Learning and LLM, and also provide explanations based on SHAP.

Chapter – 2

Review of Literature

2.1 Description of Existing Systems

Nowadays, there are a variety of healthcare systems that utilize AI technology, particularly as a means of interacting through medical chatbots. Such systems help communicate with patients, get necessary information from them, and predict possible diseases based on received input.

The majority of current models utilize such methods as machine learning and deep learning, while in recent years, some systems started applying LLM (Large Language Model). For example, some of the models apply SVM, CNN, deep learning algorithms or transformer architectures such as BERT and GPT. All such systems use datasets for training collected either from websites such as Kaggle, health forums, or medical databases.

In addition, hybrid approaches are being used to improve systems, such as LLMs together with other models like Random Forest, XGBoost, or frameworks like RAG (Retrieval-Augmented Generation).

These systems have proven their usefulness for such purposes as prediction of common illnesses, providing advice on mental health, helping people find proper specialties, and even providing some assistance with diagnoses. These applications proved their effectiveness especially during times like the COVID-19 pandemic when fast and efficient healthcare support was highly required.

However, as far as all these systems can be criticized, one can notice the lack of sufficient transparency of such systems. In most cases, their predictions strongly depend on the training dataset, and sometimes their results cannot be considered reliable. Also, the explanation of the process of obtaining the prediction result from the system is missing in the majority of cases.

To summarize, these systems provide significant advantages; however, there is always room for improvement.

2.2 Summary of Literature Study

Table [2.2.1]: Literature Review

S. No	Title	Year	Work Done	Dataset Used	Models	Conclusion	Limitations
1	Large Language Modelling and Classical AI Methods for Healthcare	2023	Explored integrating LLMs with classical AI for predictive modeling in healthcare.	NHANES dataset (2005–2010)	Random Forest, Logistic Regression, Naïve Bayes, Decision Tree	Combining LLMs with ML models improves diagnostic accuracy and predictive capabilities.	AI bias, ethical concerns, and accuracy issues in medical applications.
2	AI-Based Medical Chatbot for Disease Prediction	2024	Developed an NLP & ML-powered chatbot for disease prediction through symptom analysis.	Pre-trained datasets	NLP, ML, Deep Learning, Reinforcement Learning	AI chatbots can enhance early disease detection and healthcare accessibility.	Data privacy, security, reliability, and scalability concerns.
3	Medical Diagnosis and Treatment Recommendations (Doc-Bot) using LLMs	2024	Developed Doc-Bot, an AI assistant integrating ML & LLMs for disease prediction and treatment suggestions.	6,800 samples across 135 diseases	GPT-3, GPT-4, RoBERTa, Random Forest	Doc-Bot improves diagnostic accuracy, patient education, and clinical decision-making.	Dataset size limitations, biases in training data, and difficulty interpreting user queries.
4	AI-Based Medical Chatbot for Infectious Disease Prediction	2022	Created a deep learning & NLP-based chatbot to predict infectious diseases like COVID-19 and flu.	Disease dataset	LSTM, RNN, Decision Tree	Chatbots can improve self-diagnosis & healthcare accessibility, especially during pandemics.	Accuracy depends on training data, risk of misdiagnosis, lack of clinical validation.
5	AI-Driven Medical Chatbot for Infectious Disease Management	2025	Built an ML & NLP chatbot for symptom monitoring and real-time health guidance.	Health dataset	LSTM, RNN, BiLSTM	Enhances early disease prediction, but further improvements in reliability are needed.	Data bias, misdiagnosis risks, ethical concerns in AI-based healthcare.
6	Medical Chatbot for Infectious Disease Prediction using SVM & CNN	2024	Used SVM & CNN models to analyze symptoms and predict infectious diseases.	Kaggle dataset	SVM, CNN	Improves healthcare accessibility, particularly in remote areas.	Dataset dependency, misdiagnosis risks, ethical concerns.
7	DocBot: Integrating LLMs for Medical Assistance	2025	Developed Doc-Bot using Retrieval-Augmented Generation (RAG) for real-time medical assistance.	The Gale Encyclopedia of Medicine	RAG, Mistral.AI, NVIDIA NeMo Instruct 2407	Provides real-time, evidence-based medical support, but should be a complementary tool.	Biased training data, high computational costs, and inability to replace human doctors.
8	Chatbot for Chronic Disease Diagnosis using LLMs	2024	Introduced Chat Ella, a GPT-2-based chatbot for diagnosing chronic diseases.	Medical dataset with 24 chronic diseases	GPT-2, Adam Optimizer, CLR Scheduler	Improves remote healthcare services, particularly for patients lacking access to medical professionals.	Accuracy depends on training data, lacks clinical validation, NLP limitations.

Based on previous research, it can be stated that numerous studies were performed with an objective to develop AI-based medical chatbots to predict diseases. Multiple approaches were developed including SVM, Naïve Bayes, Random Forest, LSTM, and CNN. The use of LLMs has become quite popular lately and many researchers started implementing BERT, GPT, and LLaMA into their projects to gain more insights regarding users' symptoms. Also, many researchers decided to combine different models to achieve greater results. For instance, hybrid approach based on combining LLMs and other machine learning models was used in some cases. All these models could be trained using various datasets including Kaggle datasets, medical record data, health surveys, and others. Applications of AI-based medical chatbots are different and include infectious disease prediction, chronic disease diagnosis, basic health care support, and more.
However, it should be mentioned that all studies revealed several challenges related to the usage of such models. To be more specific, most approaches were heavily dependent on data quality. Moreover, problems such as data bias, insufficient validation, difficulties with interpreting results are typical for most approaches. Thus, although there are certain achievements, the development of more reliable and accurate solutions is required.

Most of the study mainly depends on the implementation of artificial intelligence (AI) to construct medical chatbot. The chatbot are able to interact with people, gather their symptoms, process the medical data, and make predictions about likely diseases. The advanced models such as large language models (LLMs), classical machine learning techniques, and deep learning enable them to perform so. One of them utilized deep learning and NLP methodologies, while an- other utilized SVM and CNN models on a Kaggle dataset. The chatbots were designed to predict illnesses such as flu and COVID-19.

Even though there were concerns about clinical reliability and data quality in training, these models were useful for improving healthcare accessibility, particularly during pandemics or in remote areas with poor medical infrastructure.

Different hybrid model approaches were tested, mixing LLMs with other AI techniques like Random Forest, SVM, CNN, LightGBM, XGBoost, and Retrieval- Augmented Generation (RAG). Most of

the research employed standard pre- trained models like BERT, GPTs of different versions of 2, 3.5, 4, LLaMA-2, DialoGPT, and other fine-tuned medical data models.

These models were ap- plied in a variety of areas of medicine such as the forecasting of chronic disease, infectious diseases management, mental health care, support for surgical train- ing, forecasting of medical specialties, diagnosis of eye diseases, and prescribing drugs.
Doc-Bot employed a Retrieval-Augmented Generation (RAG) framework fueled by LLMs and was trained on The Gale Encyclopedia of Medicine. Chat Ella em- ployed GPT-2 and was trained on a 24 chronic disease dataset.

Independently, KM-BERT, trained on 80,000 question-specialty pairs, performed a accuracies of 70.6% and 88.5%. The BERT-driven model greatly enhanced the task of identify- ing the appropriate medical specialty, optimizing hospital workflow and specialist allocation.

The datasets used in these tests were obtained from a variety of sources — public datasets, Kaggle, national health surveys, clinical notes, medical reviews, online Q&A forums, and hospital records. Multimodal data in the textual and medical image forms e.g., applying YOLOv7 and ResNet-50 to diagnose eye diseases were used in some studies too.

Data processing steps involved multiple times gath- ered symptoms, text preprocessing, tokenization, sentiment analysis, followed by training & test the model. The performance of models was compared against a wide performance metrics contains precision, recall, F1-score, BLEU score, per- plexity, and top-k accuracy.

These LLMs and AI-driven chatbots carry huge possibilities of enhancing health- care by making healthcare accessible and more efficient, particularly for the early diagnosis of disease. There is still work to improve quality of data and to make the models more accurate and unbiased, to ensure ethical use of that, the com- pliance with healthcare laws, and the merging of multiple sources of data so that these systems are safer and more effective on the ground.
sentiment-based drug recommendation based on Word2Vec, LSTM, LightGBM, SVM, and Naïve Bayes. In the same way, RMSCB employed a tailored ML pipeline for initial clinical assistance.

BERT-based medical chatbot was trained based on around 11,000 question- answer pairs with diverse medical datasets such as MIMIC-III, BioASQ & COVID- 19, providing dense clinical context and coverage of terminologies. Fine-tuning on this dataset, BERT surpassed classical models such as LSTM, Bi-LSTM, and SVM in processing sophisticated medical questions. Although issues of compu- tational overhead, data bias, and interpretability exist, the model's excellent performance suggests its ability to improve telemedicine, patient engagement, and healthcare assistance.

Chapter – 3

Proposed Method

3.1 Design Methodology

3.1.1 Data Collection and Data Preprocessing

In this project, a large-scale medical dataset was used, which contains around 2,46,000 records. Each record includes a set of symptoms along with the corresponding disease label. The dataset for this project was collected from reliable sources from online like kaggle and publicly available datasets, so it covers different types of diseases, including both common illnesses and some less frequent conditions of health.

Prior to training with the dataset, there were several preprocessing steps carried out. The first step involved data cleansing, which entailed removal of duplicate records and correction of any inconsistencies or missing data. Data quality issues may significantly affect the performance of a machine learning model; hence, it is essential to have clean data before proceeding with training.

Additionally, there was an encoding step aimed at converting the symptoms' textual data to numeric data, which can be easily interpreted by both machine learning and deep learning algorithms. In addition to that, the symptoms' textual data underwent natural language processing, where it was converted to lower case, punctuation removed, stop words eliminated, and lemmatization was conducted.

3.1.2 Data Splitting

In order to test and train the models appropriately, the entire dataset was split into two parts – training and testing – using an 80:20 split. About 80 percent of the dataset was used for training purposes, while the rest of the dataset was used for testing the effectiveness of the model.

The primary reason for the split is the requirement to determine the extent of the performance of the model on the new dataset. In this case, the test set was used entirely separately from the training set, without affecting its performance.

This is one of the essential steps in all machine learning tasks, as it prevents overfitting and makes sure that the model can be generalized for any user inputs.

3.1.3 Model Building and Evaluation

For the implementation of disease prediction in this research project, models of deep learning and large language models (LLMs) were applied. The deep learning models that are used include MLP, TabNet, and Residual MLP, whereas models of LLMs include XLM RoBERTa, ERNIE, and ELECTRA.

For training and evaluation of models, an extensive and processed dataset was used along with accuracy, precision, recall, and F1 score for the assessment of their efficiency in disease predictions.

From the results, it can be observed that ELECTRA performed better among the LLM models with an accuracy of around 85.42%. Among the Deep Learning models, the Multilayer Perceptron (MLP) gave the best performance with an accuracy of about 86.71%, followed closely by Residual MLP and TabNet. Overall, Deep Learning models slightly outperformed LLM-based models in terms of accuracy in this project.

These comparisons helped in identifying which model is more suitable for the system, based on both performance and reliability.

Large Language Model is built by transforming the symptom data into tex- tual format data and then it is processing by using pretrained tokenizer. The tokenized data is taken as the inputs are fed into the transformer-based model, which captures the contextual relationships through the attention mechanisms of the model. A classification method is added to perform the disease prediction. The model is trained using data and then optimized with loss functions

and optimizers. After training the data, SHAP model is integrated to which explain the predictions by assigning the importance scores to input tokens, which enable the better interpretability of the model's decisions.

Deep Learning Model is built using the structured symptom data, undergoes to the preprocessed and converted into feature vectors. The dataset is split into two categories i.e., training and testing datasets before going it into a neural network consists of input, hidden and dense layers with ReLU activation, and have a dropout layer to prevent overfitting. The output layer performs multi- class classification use the softmax activation function.

3.1.4 Prediction

After training and evaluating the models, the final step is prediction. In this phase, the system takes user input in the form of symptoms through the chatbot interface. The input is first preprocessed using tokenization and other necessary steps to convert it into a format suitable for the models.

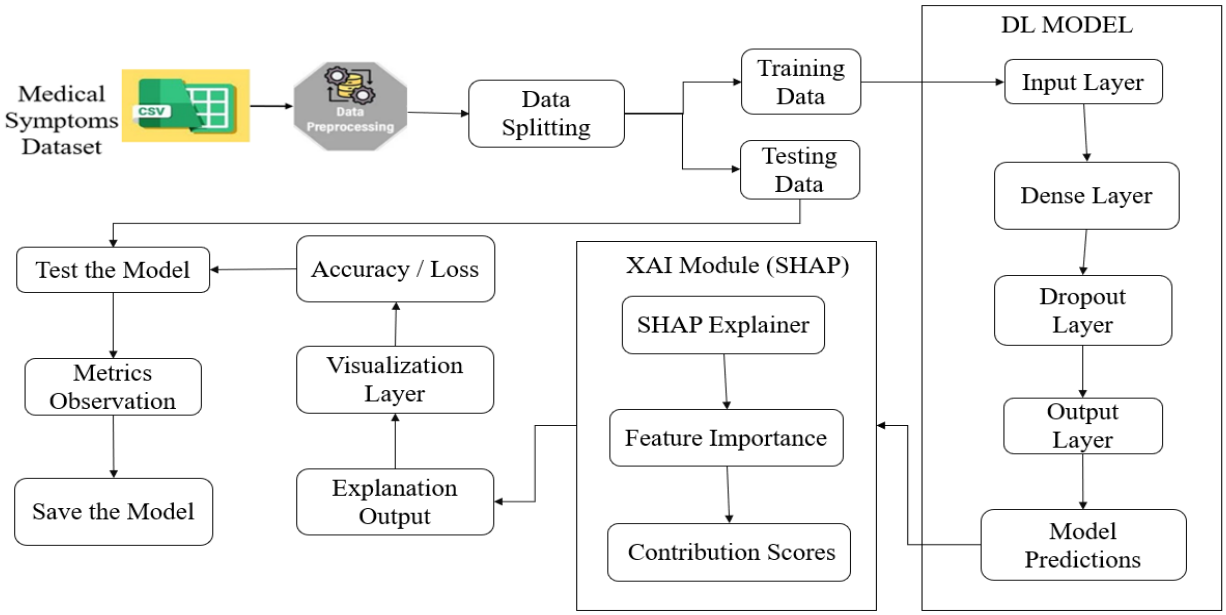
The processed input is then passed to the trained models, which analyze the symptoms and generate predictions for possible diseases. The output is given in terms of probability scores for different diseases, and the disease with the highest probability is selected as the final result.

3.1.5 Explainability (SHAP Integration)

In this project apart from prediction, it also focuses on explainability. SHAP (SHapley Additive Explanations) is used to understand how the model is making decisions. It shows which symptoms are contributing more to the predicted disease.

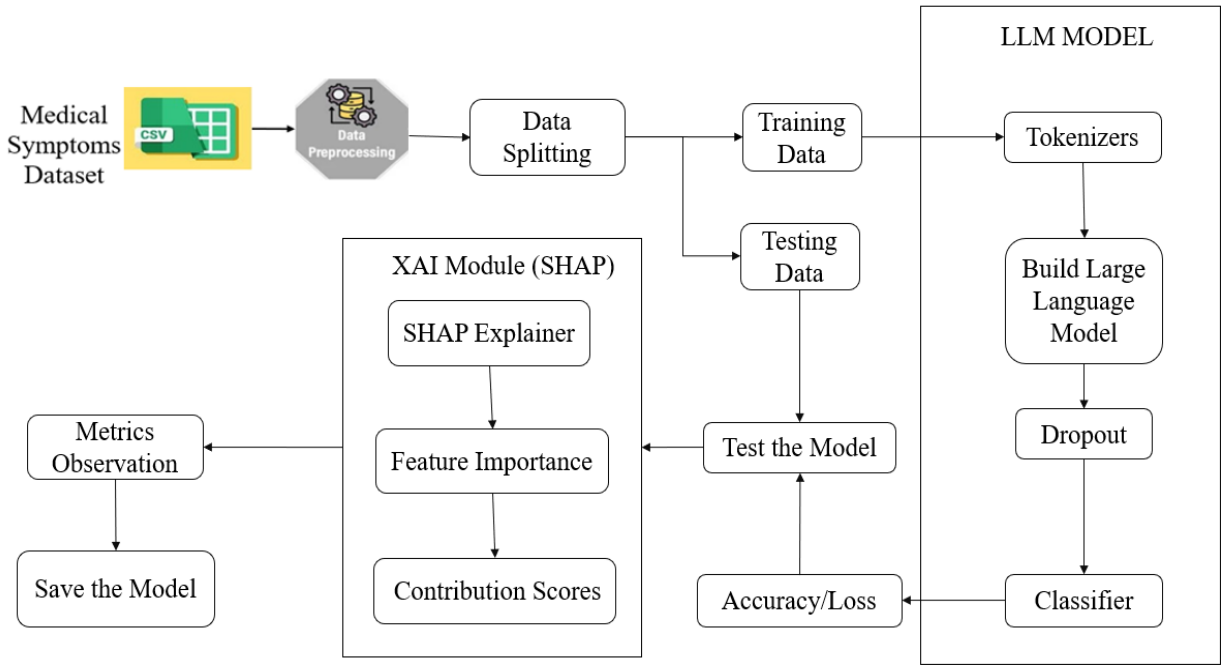
This is important because many AI models act like black boxes, where users do not know how the result is generated. By using SHAP, the system becomes more transparent and trustworthy. It also helps in better analysis of model behavior and improves confidence in the predictions.

3.2 System Architecture Diagram



Fig[3.2.1]: Architecture Diagram for Dl Models

Deep learning architecture shows the structured symptoms data, which undergoes the preprocessing and feature before split into training and test datasets. It uses the neural network undergoes to the input, hidden and output layers to learn the patterns from the training dataset with ReLU activation and having the dense layers with a dropout layer. The final output layer performs multi-class classification to predict diseases. After prediction, XAI model(shap) is applied to explain how each symptom affects the results. It helps which features are important for extraction the disease based on the symptoms**.**



Fig[3.2.2]: Architecture Diagram for LLM Models

The LLM- based architecture processes symptoms dataset by transform- ing it into the contextual format, followed by the tokenization using the pre- trained tokenizer. The LLM understands the relationships between the disease and symptoms using the classifier. The model is evaluated using the accuracy and loss. then Shap model is used to explain the predictions symptoms influenced the result.

The figure shows the overall working flow of the proposed symptom-based disease prediction system. It starts with the medical symptoms dataset, which is taken in a structured format (CSV file). This dataset contains information about different symptoms and their related diseases.

The next step in the process is data preprocessing. In this stage, the data is cleaned and prepared by removing unwanted values and converting it into a suitable format. After preprocessing, the dataset is divided into two parts: training data and testing data. The training data is used to train the models, while the testing data is used later to check the performance of the system.

Once partitioning is done, various models are employed. Both deep learning models like MLP, TabNet, and Residual MLP as well as language models like XLM-RoBERTa, ERNIE, and ELECTRA are implemented here. Training takes place to allow the algorithm to capture the relationships between the symptoms and their corresponding diseases.

The next step after training is the test phase. In the test phase, performance evaluation takes place using metrics such as accuracy and loss. The aim of this process is to determine the effectiveness of the various models through their performance.

Additionally, during the testing phase, Explainable AI employing SHAP technique is done. With this technique, the significance of each symptom in the prediction process becomes clear.

Finally, once model performance becomes satisfying, model storage takes place. At this point, the model will be stored for future use in the chatbot environment.

3.3 Description of Algorithms

3.3.1 ELECTRA

ELECTRA (Efficiently Learning an Encoder that Classifies Token Replacements Accurately) is a transformer-based machine learning model for text classification applications. Different from typical methods where the objective is to predict missing words, ELECTRA employs a unique technique called replaced token detection that makes its training process quicker and more efficient. In this study, ELECTRA is employed to categorize diseases according to symptom descriptions that are provided as text input.

Algorithm: Fine-tuning ELECTRA for Disease Classification
Input: Symptom data converted into text format
Output: Trained ELECTRA model for predicting diseases
Method:

- Prepare the symptom dataset in the form of text.
- Turn the text into tokens using ELECTRA tokenizer with padding and truncation.
- Construct a dataset and DataLoader for better data management.
- Load ELECTRA model for text sequence classification with multiple disease categories.
- Train the ELECTRA model with cross-entropy loss function and AdamW optimizer.
- Test the trained model with the test dataset and assess its performance.
- Interpret the outcomes using accuracy score, loss value, and confusion matrix.

3.3.2 ERNIE

ERNIE (Enhanced Representation through kNowledge Integration) is a pre-trained model that emphasizes understanding the semantic and contextual meaning of texts. Unlike other models such as BERT, ERNIE can extract more informative data from inputs. This study uses the ERNIE model for classifying diseases according to the symptoms presented by the user.

Algorithm: Fine-tuning ERNIE for Disease Classification
Input: Symptom descriptions in text format
Output: Trained ERNIE model for disease prediction
Method:

- Translating the symptom dataset into natural language.
- Tokenizing the text using ERNIE tokenizer.
- Loading the ERNIE model for classification tasks.
- Training the model with cross-entropy loss and AdamW optimization technique.
- Model evaluation using accuracy and other performance metrics.
- Analyzing model results using classification reports and plots.

3.3.3 XLM-RoBERTa

XLM-RoBERTa is a transformer-based model trained on many languages; hence, it can efficiently process various forms of input data. The model is mostly used for classification problems, where

one predicts the class of the input data. Here, XLM-RoBERTa will be used to predict diseases from their symptoms' descriptions.

Algorithm: Fine-tuning XLM-RoBERTa for Disease Classification

Input: Symptom data converted into text format

Output: Trained XLM-RoBERTa model for disease prediction

Method:

- Preprocess input symptoms as text descriptions.
- Convert input using XLM-RoBERTa tokenizers.
- Prepare datasets and DataLoader for training.
- Load the model for multi-class prediction.
- Train model using cross entropy and AdamW optimizer.
- Test the model using the test datasets.
- Vizualize model performance using accuracy, loss, and confusion matrix.descriptions.

3.3.4 Multilayer Perceptron (MLP)

The Multilayer Perceptron (MLP) is one of the simplest types of deep learning architectures. It comprises a network with several layers of neurons, such as an input layer, a number of hidden layers, and an output layer. The MLP is predominantly applied for solving classification problems where the algorithm recognizes patterns in the training data. For this project, the MLP is applied for disease prediction based on encoded symptom information.

Algorithm: Disease Prediction using MLP

Input: Preprocessed and encoded symptom data

Output: Trained MLP model for disease classification

Method:

- Encode symptom data into numbers.
- Design the architecture of the MLP.
- Use ReLU activation function for hidden layers.
- Specify the use of Softmax for the output layer.
- Train the MLP using cross-entropy loss and optimizer.

- Measure the accuracy and F1-score.

3.3.5 TabNet

TabNet is a special form of deep learning that focuses on tabular data. Using an attention model, the algorithm is capable of concentrating on crucial features while training. Thus, it is distinct from regular machine learning models since it is capable of automatic feature selection in learning. The following steps describe how TabNet will be applied in this work.

Algorithm: Disease Prediction using TabNet

Input: Structured symptom dataset

Output: Trained TabNet model for disease classification

Method:

- Pull together a dataset with appropriate pre-processing.
- Call for the creation of the model using appropriate parameters.
- Train the model using the training set.
- Concentrate on relevant features using the feature selection mechanism of TabNet.
- Test and evaluate the model using test data.
- Compare with other algorithms' performances.

3.3.6 Residual MLP

Residual MLP is a new type of MLP model that makes use of skip connections between layers to enhance learning efficiency by avoiding issues such as vanishing gradients. The project implements Residual MLP to enhance prediction accuracy as compared to regular MLPs.

Algorithm: Disease Prediction using Residual MLP

Input: Preprocessed and encoded symptom data

Output: Trained Residual MLP model for disease classification

Method:

- Symptom data transformation to numeric form.
- Construct layers using MLP with skip connection.
- Add activation functions, like ReLU.

- Softmax function used in the output layer for classification.
- Model trained using cross-entropy loss function and optimizer.
- Test model on testing data for evaluation.
- Comparison to standard MLP and other models.

3.4 Description of Datasets,Requirements and Tools

3.4.1 Dataset Description

For the creation of disease prediction models, we have used a large scale dataset of medical symptoms. The medical symptom dataset is composed of about 2,46,000 samples, where each sample represents a collection of symptoms and their corresponding disease labels. Such data will help us in identifying the mapping between symptoms and disease for better accuracy.

Data was taken from publicly accessible medical datasets, ensuring high reliability of data and variety of diseases in order to make it useful for learning purposes. Data consists of various types of diseases including common diseases such as infections and flu and some less common diseases as well.

This data is of mostly structured format and it involves representing symptoms in numeric value or using binary values for representing whether symptoms are present or not in a particular case. In the case of LLM-based disease detection models, this numeric data is converted into natural language text data format for making it suitable for processing by such models.

Various preprocessing steps have been carried out for preparing the dataset for machine learning purposes. Such steps include removing duplication, missing data and inconsistencies within data. Text Preprocessing methods used on the text data are lowercasing, punctuations removal, removal of stop words, and lemmatizing.

3.4.2 System Requirements

In order to execute and develop the project, hardware and software components must be provided and specified.

Hardware Requirements:

As for the hardware components, a powerful processor is required, namely Intel i5 or more. In addition, there should be at least 8 GB of RAM; however, 16 GB of RAM will be optimal. It is recommended to have 20 GB of free disk space to keep datasets and models and save generated files. It is desirable to have a graphic accelerator to accelerate the training process of deep learning models and LLMs.

Software Requirements:

As the programming language, the use of Python is required, because of its extensive use in machine learning and deep learning. In addition, the implementation can operate on any operating system like Windows or Linux. Many libraries are used depending on their functionality within the project. To handle the datasets and carry out some operations related to machine learning, such libraries as NumPy and Pandas are used, whereas Scikit-Learn library is responsible for implementing metrics for evaluations.

For deep learning algorithms, TensorFlow or PyTorch frameworks can be used. As for the NLP-related tasks, libraries such as Hugging Face Transformers or NLTK can be used. To develop the project and execute code, development environments, such as Jupyter Notebooks or Visual Studio Code, may be used.

3.4.3 Tools and Technologies Used

Multiple technologies and tools were utilized throughout the project implementation process. The Python programming language was chosen to create the solution because it is easy to learn and allows for the integration of numerous tools and libraries that are relevant for machine learning and AI projects. Pandas and NumPy are used to load and clean the data and perform numeric operations on it. These tools facilitate the manipulation of data and make the whole process more convenient.

For evaluation purposes, Scikit-learn can be used as an evaluation metric. It provides multiple options, including accuracy, precision, recall, and F1-score. In addition, Scikit-learn can be applied to carry out some preprocessing and model evaluation activities. Frameworks like TensorFlow or PyTorch are utilized when it comes to creating complex neural networks.

For models based on LLMs, the Hugging Face Transformer library can be used as it contains multiple pre-trained models (e.g., XLM-RoBERTa, ELECTRA, and ERNIE). These models can be fine-tuned for the task at hand. As a result, the creation of a custom one becomes unnecessary.

When it comes to explainability tools, SHAP can be used. Its goal is to analyze the internal functioning of a predictive model to identify which symptoms are crucial for a certain disease prediction.

Finally, for visualization, Matplotlib and Seaborn can be utilized to plot graphs with accuracy, loss, and confusion matrix values.

Chapter – 4

Results and Observations

4.1 Stepwise description of Results

Confusion Matrix

The confusion matrix is a performance measure that can be used for classification. It shows how good a model is at making predictions when compared to the real-life scenarios. It gives an overview of the correct and wrong predictions made by the model.

Performance Metrics

Precision

Precision refers to the number of true positives divided by the total number of true positives and false positives. It measures the accuracy of positive predictions.

$$Precision = \frac{True\ positive}{true\ \ positives + false\ \ positives} \qquad \text{-(Eq 1)}$$

Accuracy

Accuracy refers to the number of true positives and negatives divided by the total number of positive and negative predictions. It gives an insight into the proportion of accurate predictions made by the model.

$$\text{Accuracy} = \text{True positives} + \text{True Negatives/Total Instances} \qquad \text{-(Eq 2)}$$

Recall

Recall refers to the number of true positives divided by the total number of actual positives. It measures the proportion of correct positive predictions.

$$Recall = True\ Positives + False\ NegativesTrue\ Positives \qquad (Eq\ 3)$$

F1 score: F1-Score is the harmonic mean of Precision and Recall.

$$F1 = Precision + Recall/2 \cdot Precision \cdot Recall \qquad (Eq\ 4)$$

4.1.1 Electra Model

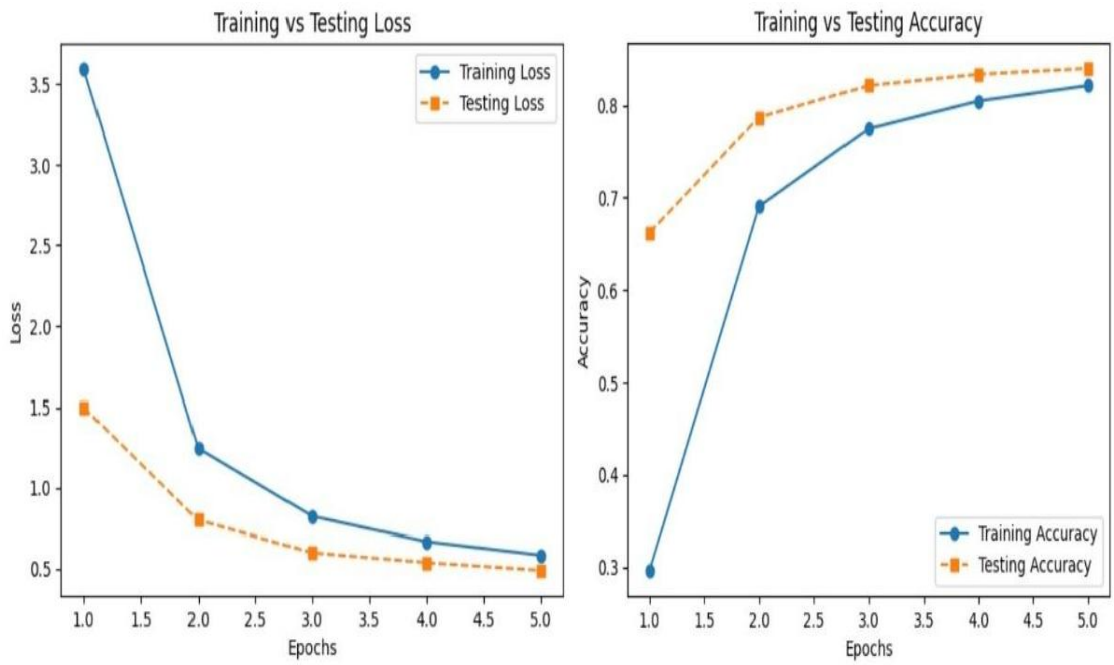


Fig1 : 4.1.1 Graphs for Electra Model

Accuracy plot depicts the fact that both training and testing accuracies start off by rising sharply at the beginning of training but soon stabilize to an average of 84%. Testing accuracy is slightly less than training accuracy, but the difference between the two is negligible.

From the loss graph, it can be seen that both training and testing loss decrease sharply at the beginning and then slowly level off at lower values. The testing loss is slightly lower than the training loss, which indicates that the model is not overfitting. Overall, the model performs well even on new or unseen data.

Table1 4.1.1 Sample Classification Report for ELECTRA Model

Disease	Precision	Recall	F1-Score	Support
Abdominal aortic aneurysm	0.91	1	0.95	20
Abdominal hernia	0.93	0.95	0.94	81
Abscess of nose	0.86	0.84	0.85	58
Abscess of the lung	1	0.83	0.91	6
Abscess of the pharynx	0.87	0.95	0.91	63
Acanthosis nigricans	0.86	1	0.92	6
Acariasis	1	0.8	0.89	5
Achalasia	0.67	1	0.8	20
Acne	0.85	0.73	0.79	103
Actinic keratosis	0.93	0.78	0.85	179
Acute bronchiolitis	0.92	0.92	0.92	253
Acute bronchitis	0.83	0.67	0.74	221
Acute bronchospasm	0.7	0.82	0.76	187
Acute fatty liver of pregnancy (AFLP)	0	0	0	3
Acute glaucoma	0.55	0.55	0.55	31
Acute kidney injury	0.97	0.94	0.95	209

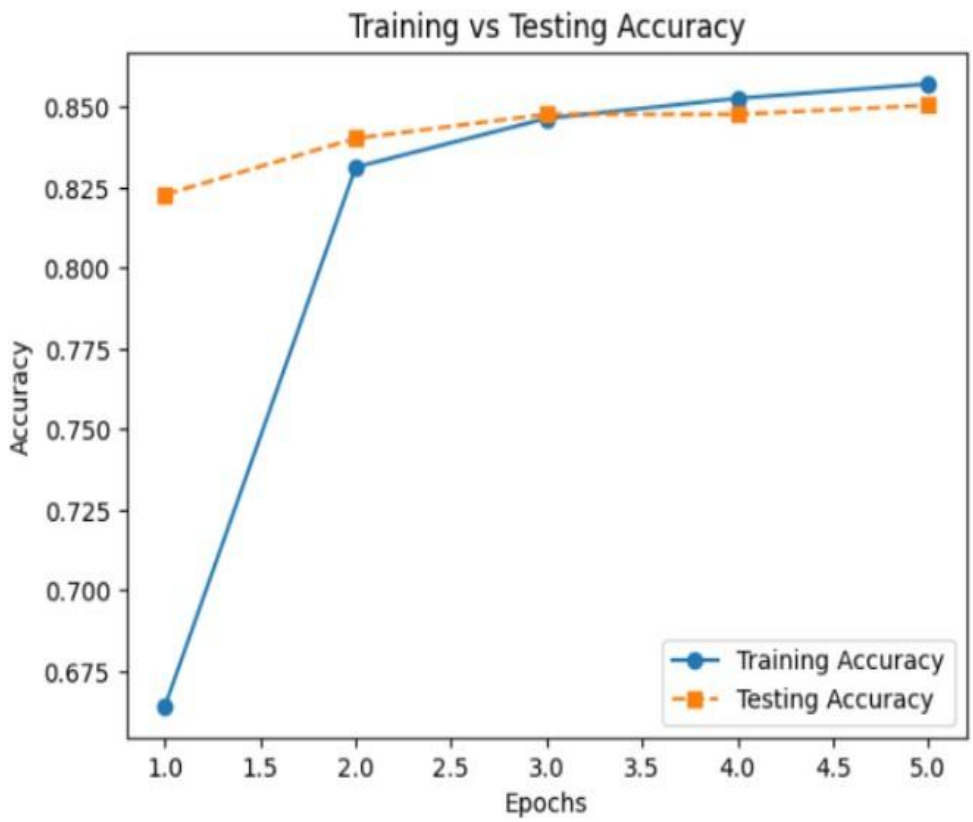
The above mentioned table is one of the sample table which shows the classification report for some diseases that are classified using ELECTRA model.

Table2 Classification Metrics for ELECTRA Model

Metrics	Precisio	Recall	F1-Scor	Suppor
Accuracy			0.84	49389
macro avg	0.76	0.73	0.73	49389
weighted avg	0.86	0.84	0.84	49389

The above table describes classification metrics for ELECTRA Model which shows the accuracy, precision, recall, fi-score, support. And from the above table we can see that the accuracy for ELECTRA Model is obtained as 84%.

4.1.2 Ernie Model



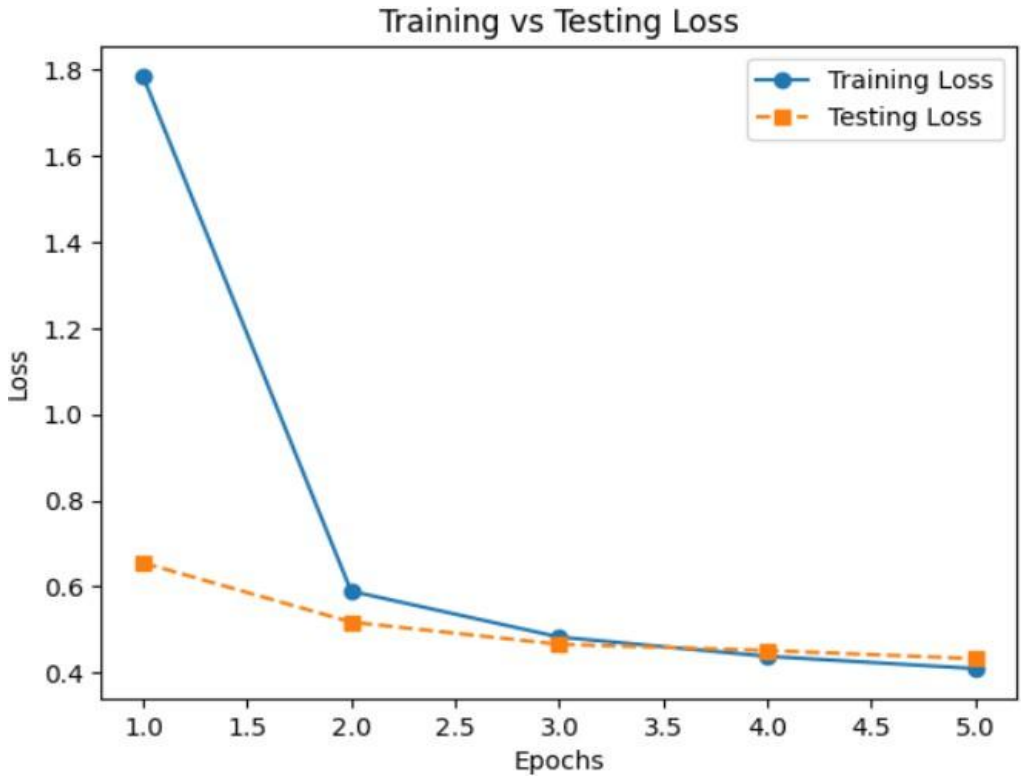


Fig1 4.1.2 Graphs ERINE Model

The accuracy graph shows that both training and testing accuracy improve rapidly in the initial epochs and then stabilize around 85%. The testing accuracy stays slightly below the training accuracy but remains close throughout. The loss graph shows that both training and testing loss decrease sharply in the beginning and then level off at low values.

Table1 4.1.2 Sample Classification Report for ERINE Model

Disease	Precision	Recall	F1-Score	Support
Amblyopia	0.93	1	0.96	13
Amyloidosis	0	0	0	1
Amyotrophic lateral sclerosis (ALS)	1	0.89	0.94	9
Anal fissure	0.97	0.94	0.96	36
Anal fistula	1	1	1	1
Anemia	0.98	0.95	0.96	57
Anemia due to chronic kidney disease	1	0.83	0.91	6
Anemia due to malignancy	1	1	1	1
Anemia of chronic disease	1	0.86	0.92	7
Angina	0.91	0.92	0.92	179
Ankylosing spondylitis	0.84	0.86	0.85	126
Anxiety	0.97	0.92	0.94	236
Aortic valve disease	0.9	0.95	0.92	84
Aphakia	0	0	0	1
Aphthous ulcer	1	0.84	0.91	44
Aplastic anemia	1	1	1	16

The above mentioned table is one of the sample table which shows the classification report for some diseases that are classified using ERINE model.

Table2 4.1.2 Classification Metrics for ERINE Model

Metrics	Precisio	Recall	F1-Scor	Suppor
Accuracy			0.85	49389
macro avg	0.8	0.79	0.79	49389
weighted avg	0.86	0.85	0.85	49389

The above table describes classification metrics for ERINE Model which shows the accuracy, precision, recall, fi-score, support. And from the above table we can see that the accuracy for ERINE Model is obtained as 85%.

4.1.3 XLM-Roberta

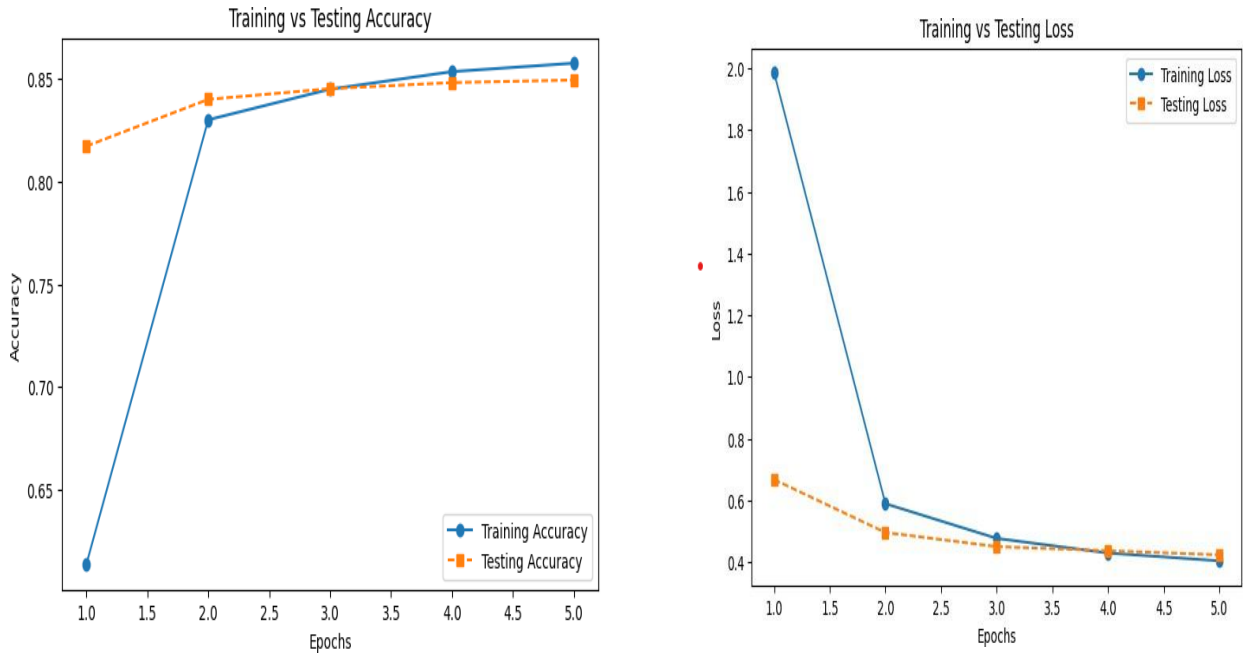


Fig1- 4.1.3 Graphs XLM-ROBERTA Model

The accuracy graph shows that both training and testing accuracy improve rapidly in the initial epochs and then stabilize around 85%. The testing accuracy stays slightly below the training accuracy but remains close throughout. The loss graph shows that both training and testing loss decrease sharply in the beginning and then level off at low values. The testing loss remains slightly lower than the training loss, suggesting that the model is not overfitting and maintains good performance on new data.

Table1 4.1.3 Sample Classification Report for XLM-ROBERTA Model

Disease	Precision	Recall	F1-Score	Support
Appendicitis	0.84	0.79	0.81	183
Arrhythmia	0.92	0.69	0.79	52
Arthritis of the hip	0.96	0.96	0.96	228
Ascending cholangitis	1	0.9	0.95	10
Asperger syndrome	0.9	0.86	0.88	80
Aspergillosis	0.75	1	0.86	3
Asthma	0.86	0.7	0.77	189
Astigmatism	0.68	0.58	0.63	55
Atelectasis	0.77	0.45	0.57	22
Athlete's foot	0.71	0.86	0.78	73
Atonic bladder	1	0.73	0.85	15
Atrial fibrillation	0.78	0.37	0.5	19
Atrial flutter	1	0.86	0.93	51
Atrophic skin condition	0.71	0.78	0.74	85
Atrophic vaginitis	1	0.74	0.85	66
Atrophy of the corpus cavernosum	1	0.6	0.75	5
Attention deficit hyperactivity disorder (ADHD)	0.7	0.94	0.8	99

The above mentioned table is one of the sample table which shows the classification report for some diseases that are classified using XLM-ROBERTA model.

Table2 4.1.3 Classification Metrics for XLM-ROBERTA Model

Metrics	Precisio	Recall	F1-Scor	Suppor
Accuracy			0.85	49389
macro avg	0.8	0.79	0.79	49389
weighted avg	0.86	0.85	0.85	49389

The above table describes classification metrics for XLM-ROBERTA Model which shows the accuracy, precision, recall, fi-score, support. And from the above table we can see that the accuracy for XLM-ROBERTA Model is obtained as 85%.

4.1.4 XLNet + ELECTRA MODEL

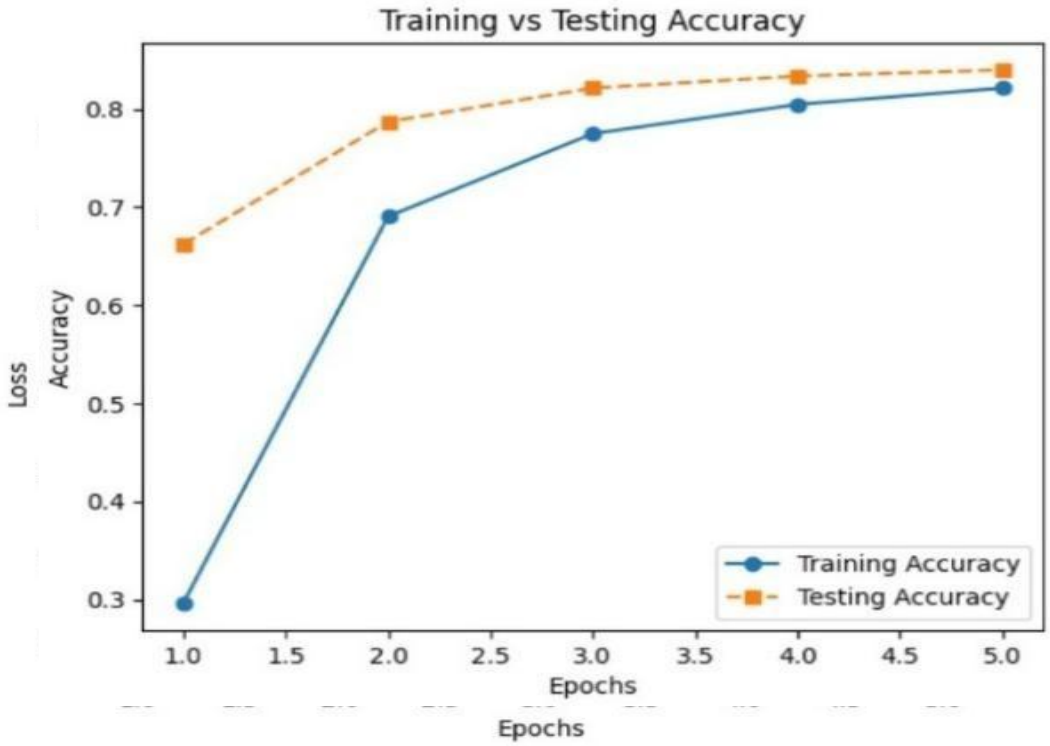


Figure1 - 4.1.4 Graphs for XLNet + ELECTRA Model

The accuracy graph shows that both training and testing accuracy improve rapidly in the initial epochs and then stabilize around 84%. The testing accuracy stays slightly below the training accuracy but remains close throughout. The loss graph shows that both training and testing loss decrease sharply in the beginning and then level off at low values. The testing loss remains slightly lower than the training loss, suggesting that the model is not overfitting and maintains good performance on new data.

Table1 - 4.1.4 Sample Classification Report for XLNet + ELECTRA Model

Disease	Precision	Recall	F1-Score	Support
Abdominal aortic aneurysm	0.91	1	0.95	20
Abdominal hernia	0.93	0.95	0.94	81
Abscess of nose	0.86	0.84	0.85	58
Abscess of the lung	1	0.83	0.91	6
Abscess of the pharynx	0.87	0.95	0.91	63
Acanthosis nigricans	0.86	1	0.92	6
Acariasis	1	0.8	0.89	5
Achalasia	0.67	1	0.8	20
Acne	0.85	0.73	0.79	103
Actinic keratosis	0.93	0.78	0.85	179
Acute bronchiolitis	0.92	0.92	0.92	253
Acute bronchitis	0.83	0.67	0.74	221
Acute bronchospasm	0.7	0.82	0.76	187
Acute fatty liver of pregnancy (AFLP)	0	0	0	3
Acute glaucoma	0.55	0.55	0.55	31
Acute kidney injury	0.97	0.94	0.95	209

The above-mentioned table is one of the sample table which shows the classification report for some diseases that are classified using XLNet + ELECTRA model.

Table2 - 4.1.4 Classification Metrics for XLNet + ELECTRA Model

Metrics	Precisio	Recall	F1-Scor	Suppor
Accuracy			0.84	49389
macro avg	0.76	0.73	0.73	49389
weighted avg	0.86	0.84	0.84	49389

The above table describes classification metrics for ELECTRA Model which shows the accuracy, precision, recall, fi-score, support. And from the above table we can see that the accuracy for ELECTRA Model is obtained as 84.57%.

4.1.5 XLM-RoBERTa + ERNIE MODEL

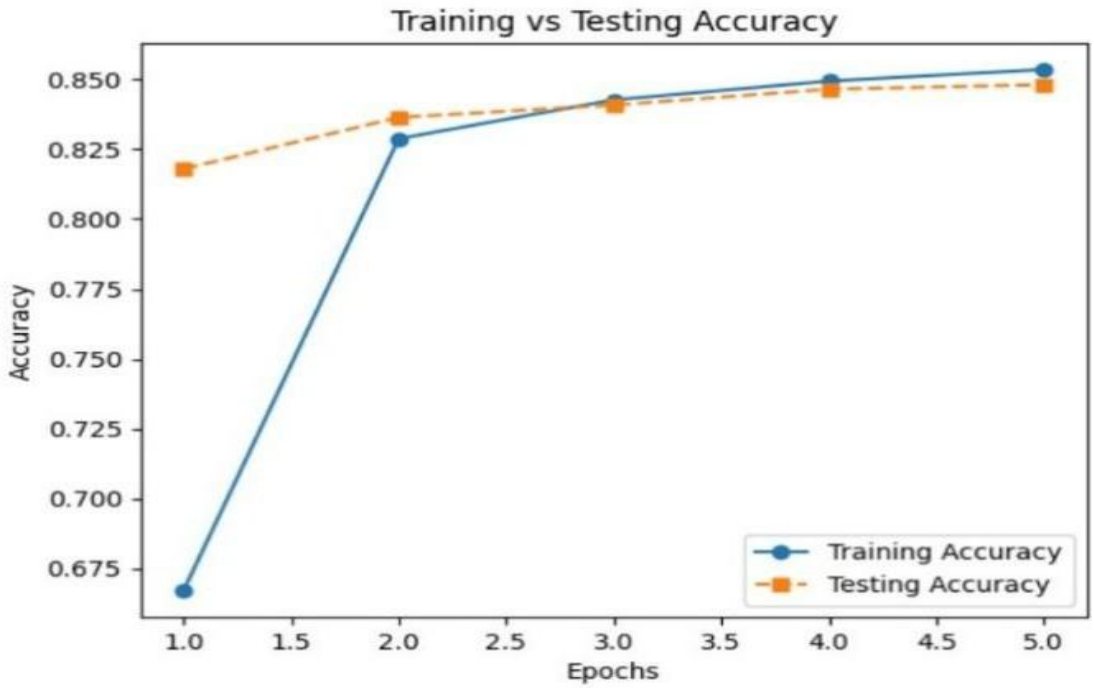


Figure1 - 4.1.5 Graphs XLM-RoBERTa + ERNIE Model

The accuracy graph shows that both training and testing accuracy improve rapidly in the initial epochs and then stabilize around 84%. The testing accuracy stays slightly below the training accuracy but remains close throughout. The loss graph shows that both training and testing loss decrease sharply in the beginning and then level off at low values. The testing loss remains slightly lower than the training loss, suggesting that the model is not overfitting and maintains good performance on new data.

Table1 - 4.1.5 Sample Classification Report for XLM-RoBERTa + ERNIE Model

Disease	Precision	Recall	F1-Score	Support
Acute otitis media	0.72	0.9	0.8	145
Acute pancreatitis	0.86	0.91	0.88	246
Acute respiratory distress syndrome (ARDS)	0.86	0.94	0.9	82
Acute sinusitis	0.8	0.83	0.81	179
Acute stress reaction	0.88	0.91	0.89	185
Adhesive capsulitis of the shoulder	1	0.82	0.9	17
Adjustment reaction	0.78	0.74	0.76	89
Adrenal adenoma	0.73	0.89	0.8	9
Adrenal cancer	0	0	0	1
Alcohol abuse	0.79	0.82	0.8	87
Alcohol intoxication	0.83	0.94	0.88	62
Alcohol withdrawal	0.91	0.97	0.93	148
Alcoholic liver disease	0.95	0.93	0.94	44
Allergy	0.76	0.96	0.84	182
Allergy to animals	1	1	1	8
Alopecia	0.77	0.83	0.8	70
Alzheimer disease	0.94	0.94	0.94	70

The above-mentioned table is one of the sample table which shows the classification report for some diseases that are classified using XLNET model.

Table2 - 4.1.5 Classification Metrics for XLM-RoBERTa + ERNIE Model

Metrics	Precisio	Recall	F1-Scor	Suppor
Accuracy			0.85	49389
macro avg	0.81	0.79	0.79	49389
weighted avg	0.86	0.85	0.85	49389

The above table describes classification metrics for XLM-RoBERTa + ERNIE Model which shows the accuracy, precision, recall, fi-score, support. And from the above table we can see that the accuracy for XLM-RoBERTa + ERNIE Model is obtained as 85%.

4.1.6 ELECTRA + ERNIE MODEL

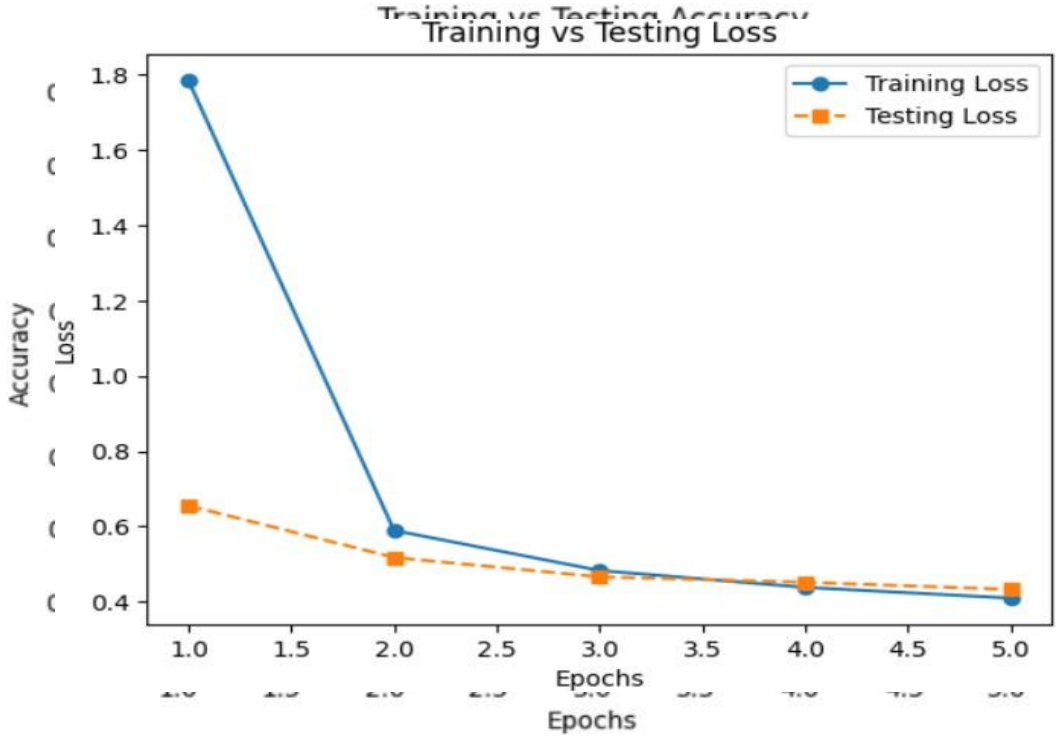


Figure-4.1.6 Graphs ELECTRA + ERNIE Model

The accuracy graph shows that both training and testing accuracy improve rapidly in the initial epochs and then stabilize around 85%. The testing accuracy stays slightly below the training accuracy but remains close throughout. The loss graph shows that both training and testing loss decrease sharply in the beginning and then level off at low values. The testing loss remains slightly lower than the training loss, suggesting that the model is not overfitting and maintains good performance on new data.

Table1 4.1.6 Sample Classification Report for ELECTRA + ERNIE Model

Disease	Precision	Recall	F1-Score	Support
Amblyopia	0.93	1	0.96	13
Amyloidosis	0	0	0	1
Amyotrophic lateral sclerosis (ALS)	1	0.89	0.94	9
Anal fissure	0.97	0.94	0.96	36
Anal fistula	1	1	1	1
Anemia	0.98	0.95	0.96	57
Anemia due to chronic kidney disease	1	0.83	0.91	6
Anemia due to malignancy	1	1	1	1
Anemia of chronic disease	1	0.86	0.92	7
Angina	0.91	0.92	0.92	179
Ankylosing spondylitis	0.84	0.86	0.85	126
Anxiety	0.97	0.92	0.94	236
Aortic valve disease	0.9	0.95	0.92	84
Aphakia	0	0	0	1
Aphthous ulcer	1	0.84	0.91	44
Aplastic anemia	1	1	1	16

The above-mentioned table is one of the sample tables which shows the classification report for some diseases that are classified using ERINE model.

Table 2 – 4.1.6 Classification Metrics for ELECTRA + ERNIE Model

Metrics	Precisio	Recall	F1-Scor	Suppor
Accuracy			0.85	49389
macro avg	0.8	0.79	0.79	49389
weighted avg	0.86	0.85	0.85	49389

When comparing all hybrid models, it is clear that the XLNet + ELECTRA combination performs slightly better than the others.

Overall, the results suggest that hybrid models can improve performance compared to individual models, but the improvement is not very large. Still, they provide a good balance between accuracy and robustness, making them useful for building reliable prediction systems.

The accuracy of the symptom-based diseases on differ- ent model with the same dataset.
The performances of three DL models such as multilayer perception (MLP), TabNet and Residual MLP along with the LLMs such as XLM-Roberta, ERNIE, ELECTRA on the classification of disease performance compared to metrics is discussed.

the accuracy of the symptom-based diseases on differ- ent model with the same dataset. The performances of three DL models such as multilayer perception (MLP), TabNet and Residual MLP along with the LLMs such as XLM-Roberta, ERNIE, ELECTRA on the classification of disease per- formance compared to metrics is discussed.

4.1.7 Deep Learning Models

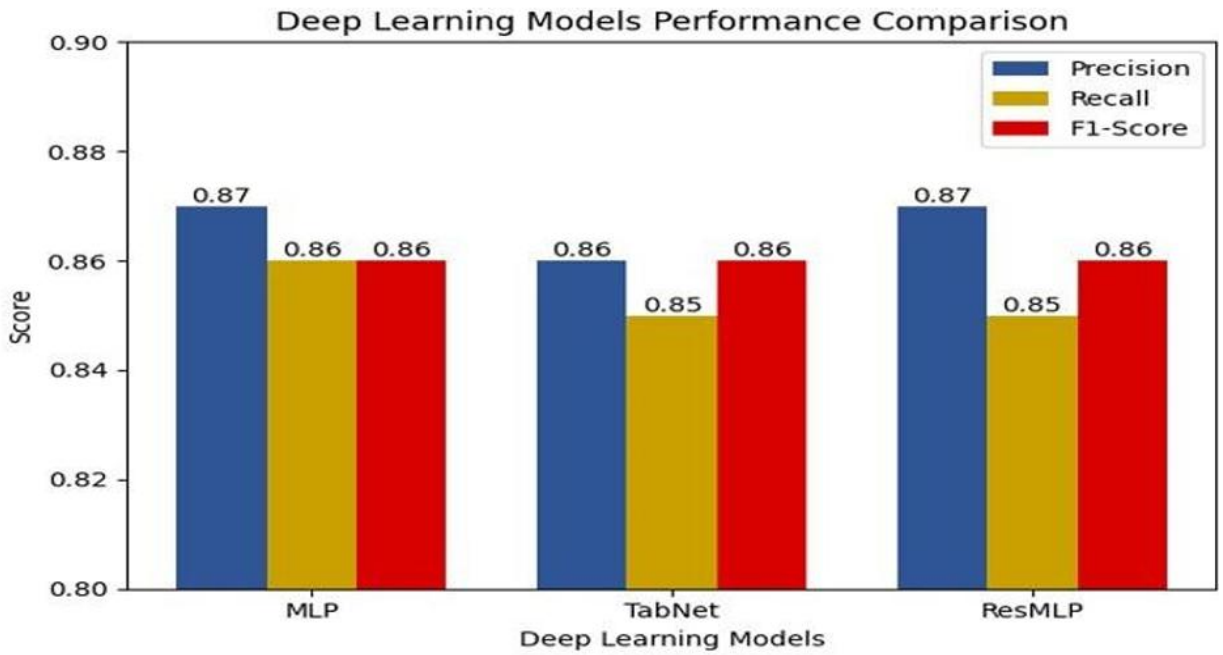


Figure-4.1.7 Graphs of deep learning models

The figure shows the performance comparison of different deep learning models, namely MLP, TabNet, and Residual MLP, based on metrics such as precision, recall, and F1-score. From the graph, it can be observed that all three models perform quite closely, with values mostly ranging between 0.85 and 0.87.

Among the models, MLP and Residual MLP show slightly better performance in terms of precision, both reaching around 0.87. TabNet, on the other hand, has a slightly lower precision of around 0.86. When considering recall, MLP performs the best with a value of 0.86, while TabNet and Residual MLP both have a recall of around 0.85.

In terms of F1-score, all three models show almost similar performance, with values close to 0.86. This indicates that the models maintain a good balance between precision and recall. Overall, the results suggest that MLP and Residual MLP perform slightly better compared to TabNet, but the difference is not very large.

4.1.8 Large Language Models

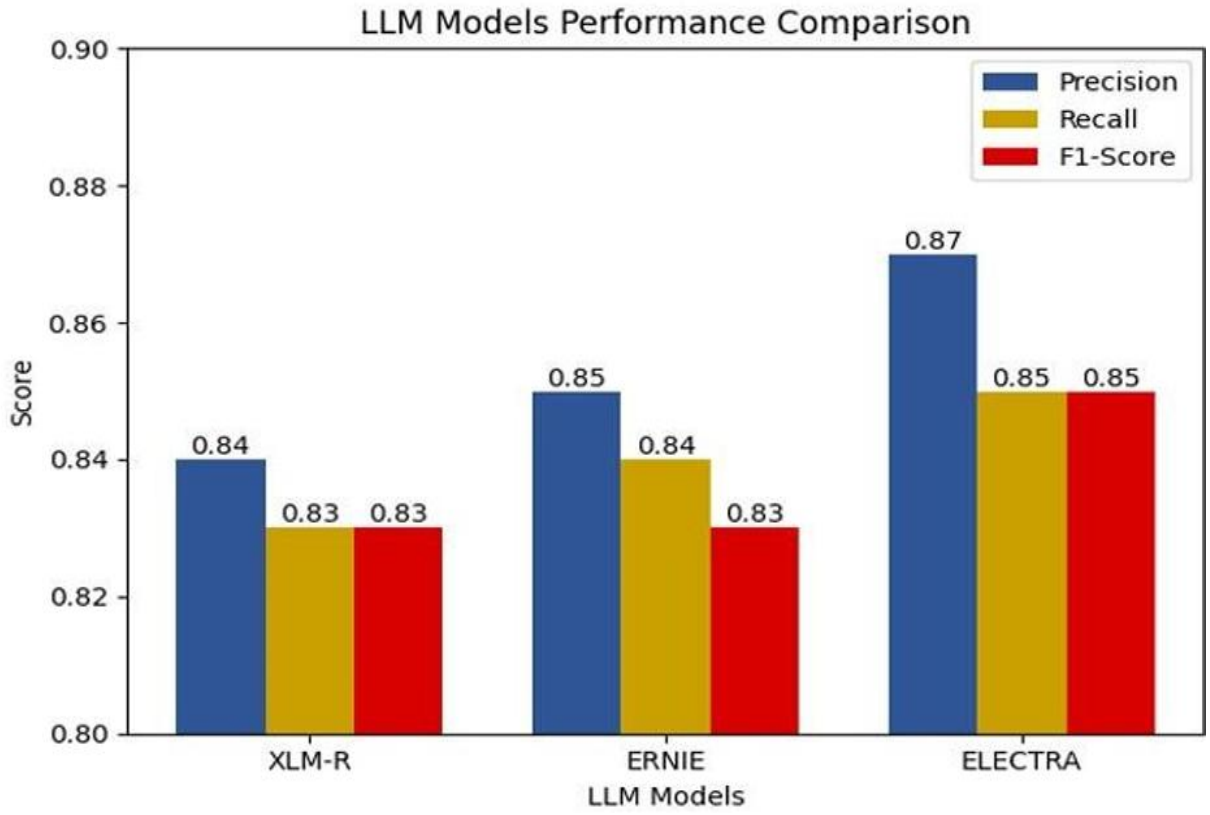


Figure-4.1.8 Graphs of large language models

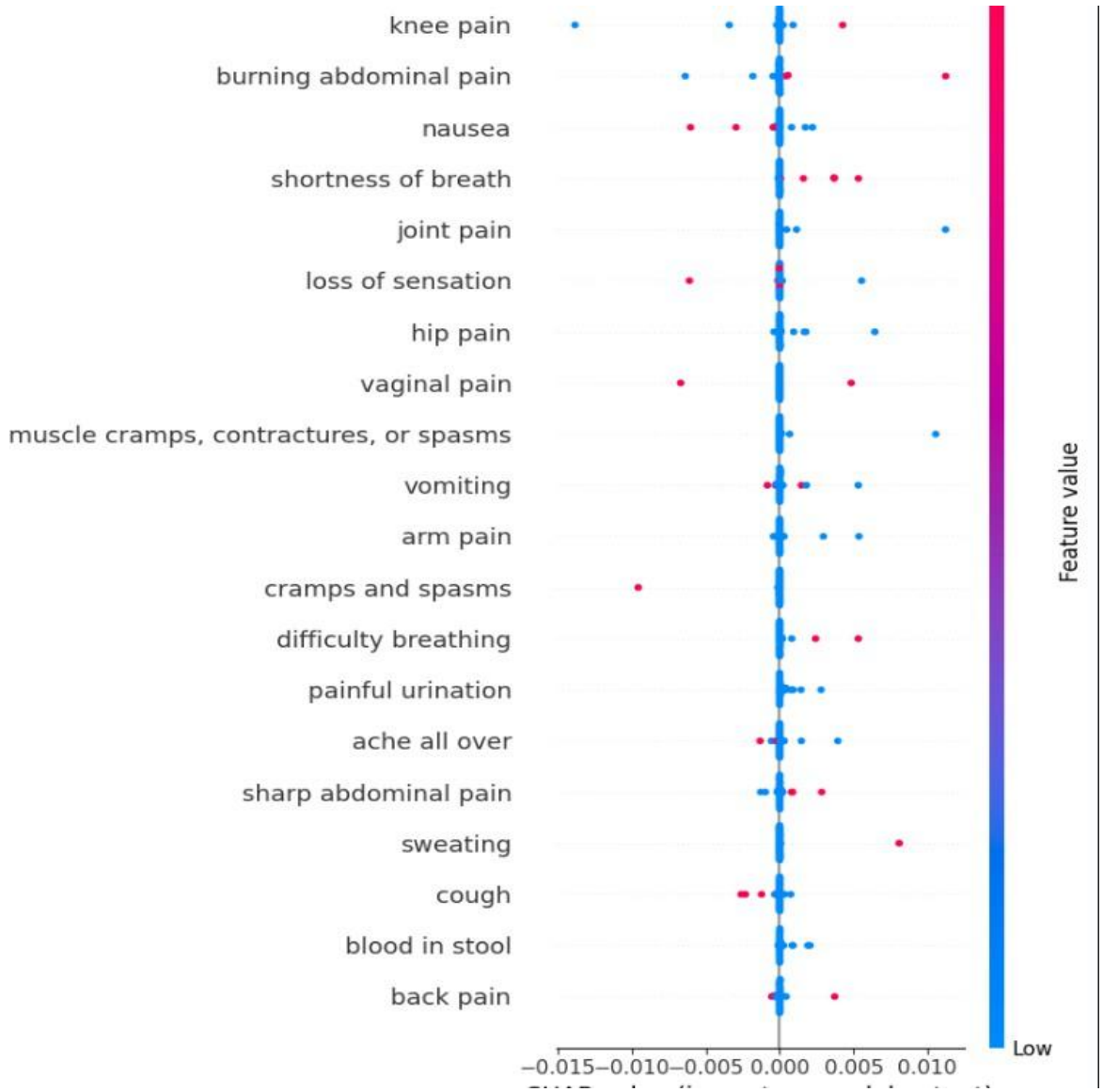
The figure shows the performance comparison of LLM-based models, namely XLM-RoBERTa, ERNIE, and ELECTRA, based on precision, recall, and F1-score. From the graph, it can be seen that all models perform reasonably well, with values mostly ranging between 0.83 and 0.87.

Among the three models, ELECTRA performs the best overall. It achieves the highest precision of around 0.87 and also shows better recall and F1-score, both around 0.85. ERNIE comes next, with

precision close to 0.85 and slightly lower recall and F1-score values. XLM-RoBERTa has the lowest performance among the three, with precision around 0.84 and recall and F1-score around 0.83.

Overall, the results indicate that ELECTRA provides better performance compared to the other LLM models in terms of all three metrics. However, the difference between the models is not very large, which shows that all of them are capable of handling the disease prediction task effectively.

XAI Analysis approach



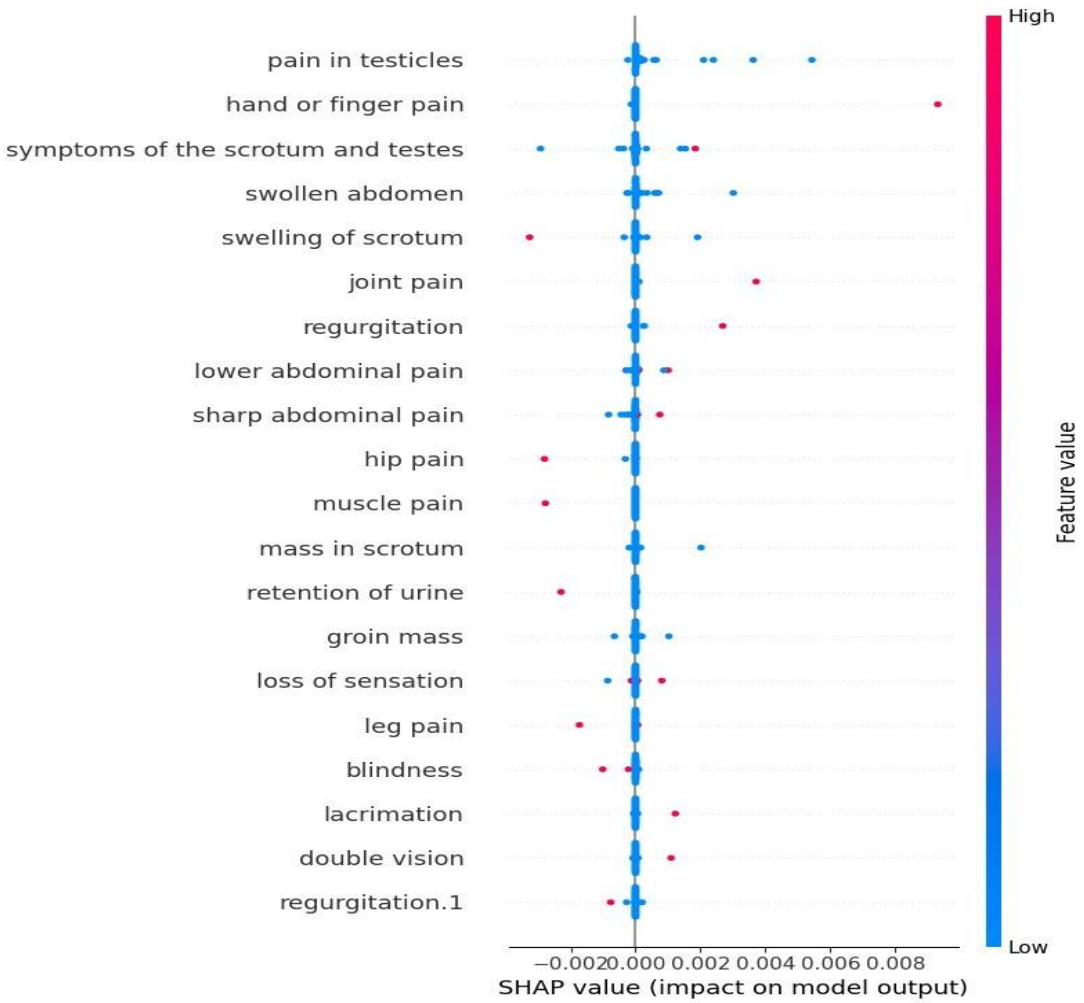


Fig 4.1.9: Explainable AI (XAI) analysis result for the deep learning model.

An Explainable Artificial Intelligence (XAI) approach was employed using SHAP (SHapley Additive Explanations). SHAP is based on game theory and provides a unified measure of feature importance by estimating the contribution of each input feature to the model's prediction.

The XAI(SHAP) model summary the plots illustrate the based on the impact of different types of symp- toms on the model's prediction. Features or symptoms such as knee pain, ab- dominal pain, nausea, and joint pain etc shows higher influence, as indicated by a wider spread of SHAP values. Similarly, symptoms like pain in testicles, swelling, and

urinary-related issues play a significant role in specific class pre- dictions. The colour gradient (blue to red) represents low to high feature value, while the horizontal distribution indicates the strength and direction of impact.

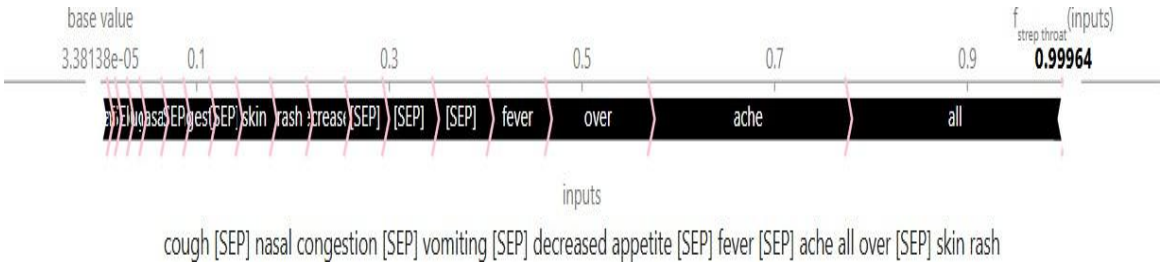


Fig-4.1.10 : Explainable AI (XAI) analysis results for Large Language model

A SHAP model plot the explaining of the individual prediction made by LLM model. The model predicts the disease class with a high score of 0.99964 for the predicition the sample disease from the test dataset, influenced by multiple symptoms such as cough, nasal congestion, vomiting, decreased appetite, fever, headache, and skin rash. The image shows that how each feature contributes to pushing the prediction toward the final output, with the arrows indicating the direction along with the strength of influence.

4.2 Test case results/Result Analysis

Testing was conducted on the proposed algorithm to see how the performance will be in the process of implementing the disease-prediction algorithm based on the presented symptoms. Thus, the main goal of the research will be the evaluation of model efficiency in predicting diseases according to their symptoms. For this reason, several models, including the Large Language Model, hybrid model, and Deep Learning models were used in the research.

The quality of testing models was estimated according to the following metrics: accuracy, precision, recall, and F1-score. Each of these metrics reflects a different aspect of the model's performance. For instance, accuracy can give general information about how many predictions the algorithm makes correctly, while precision can estimate positive predictions. Recall estimates positive predictions as well, but it looks at true positives. Finally, F1-score combines precision and recall in one metric and provides better evaluation of classification models.

The majority of models tested demonstrate quite a similar behavior during training. First of all, it can be said that at the very beginning of training, the accuracy values for both training and testing rise rapidly. Then, after a couple of epochs, the values stop rising and become stable. It means that the model has successfully learned the main characteristics and patterns and does not require any additional training. Moreover, the difference between training and testing accuracy values is quite insignificant, which means that the overfitting problem is avoided.

The loss graphs confirm the observations above. Namely, at the beginning of training, the loss values for both training and testing processes go down rapidly. Then, after a certain period, loss values remain stable and rather low. Most often, testing loss values were slightly lower compared to the training loss values. Therefore, one may conclude that training goes quite efficiently.

Talking about the models based on LLM technology, one should pay attention to the following fact. All of these models show pretty decent performance; however, they differ in terms of accuracy value. Namely, the accuracy value for ELECTRA model was about 84% and remained steady. ERNIE showed slightly better results since its accuracy reached 85%. XLM-RoBERTa demonstrated almost the same results by reaching accuracy of 85%. These models are especially valuable due to natural language processing abilities.

Furthermore, experiments on hybrid models were performed to check if the combination of architectures improves the result. The best result among them is demonstrated by the model XLNet + ELECTRA, having the accuracy of around 84.57% with stable predictions. Also, models XLM-RoBERTa + ERNIE and ELECTRA + ERNIE achieved accuracy close to 85%. Therefore,

according to the results, hybrid models allow for slightly improving the prediction performance using characteristics of different models.

Another interesting observation from the reports of classifications is the ability of the models to ensure a good balance between precision and recall. It means that besides being precise, the models have high chances of identifying correctly the disease classes. Moreover, the value of support in the reports demonstrates that there was enough of samples for each class, which allows for more reliable predictions.

Thus, when analyzing deep learning models, one should say that there is almost no difference between MLP, TabNet, and Residual MLP, as the values for precision, recall, and F1-score are mostly between 0.85 and 0.87. Between the mentioned models, MLP and Residual MLP perform slightly better regarding the values of accuracy and precision, while TabNet shows relatively low but still quite decent results. In other words, traditional deep learning is still efficient in dealing with structured symptoms' data.

The same tendency can be noticed for LLM-based models, as the best result here is obtained by ELECTRA regarding precision and overall performance. However, the difference in this case is not that noticeable, meaning that all of them are efficient in solving the problem.

An important thing observed during testing was that all models showed consistent performance regardless of the kind of symptom input, which demonstrates a high level of adaptability of the system under analysis.

Finally, to summarize, the presented research proves the efficiency and robustness of the proposed system for predicting diseases. Using deep learning, language modeling, and hybrid models allowed for obtaining a powerful prediction system. Additionally, introducing the explainability approach (SHAP) made the system more reliable for real-world applications.

To conclude, test-case analysis demonstrates the efficiency of the models used for predicting diseases.

Table1 - 4.2 Result Analysis of all the Models for mini project 1

MODEL	TEST ACCURACY
ELECTRA	**84%**
ERINE	**85%**
XLM-ROBERTA	**85%**

The accuracy results of different LLMs models utilized in this study prove the ability of the tested models to predict diseases with high efficiency. Thus, the ELECTRA model demonstrated 84% accuracy in the task, while the ERNIE and XLM-RoBERTa models had a slight edge with their 85% accuracy. Although the results are quite close to each other, they clearly indicate the efficiency of the tested models.

As for the ELECTRA model, the performance of the model is quite stable, with an accuracy of 84%. The strength of this model lies in the fast convergence rate and effective training compared to other models based on the transformers approach. The model has shown the ability to learn relevant information from the symptoms and make accurate predictions. However, the accuracy score remains lower than in the case with other models.

The above table describes classification metrics for ELECTRA + ERNIE Model which shows the accuracy, precision, recall, fi-score, support. And from the above table we can see that the accuracy for ELECTRA + ERNIE Model is obtained as 85%.

Table2 - 4.2 Result Analysis for all Models for mini project 2

Hybrid MODEL	TEST ACCURACY
XLM-roberta + ERNIE	84.69%
ELECTRA + ERNIE	84.75%
XLNet + ELECTRA	85.17%

The hybrid models used in this project combine two different transformer architectures to improve the overall performance of disease prediction. From the results, it can be observed that all hybrid models perform well, with accuracy values ranging between 84.69% and 85.17%. This shows that combining models is a useful approach, as it helps in capturing different aspects of the input data.

At the same time, XLM-RoBERTa + ERNIE has proven itself successfully, reaching the accuracy of about 84.69%. The mentioned approach involves the application of such hybrid models, which will combine the multilingual understanding capability of XLM-RoBERTa and knowledge integration capability of ERNIE. The described model demonstrates relatively good results; however, the accuracy level is significantly lower than in another hybrid models. It can be explained by the fact that the mentioned models involve deep contextual analysis; thus, it provides less advantage during analysis of structured information.

The best performance among all the models discussed above is demonstrated by ELECTRA + ERNIE, whose accuracy reaches 84.75%. At the same time, one should note that the strengths of ELECTRA are associated with an effective learning process and fast convergence of information, whereas ERNIE demonstrates an advantage in analyzing semantic relationships. Nevertheless, despite the better performance compared to its precursor, the proposed approach cannot be regarded as the optimal one in the scope of the current research.

Finally, the XLNet + ELECTRA approach is the most effective one; in particular, it achieves an accuracy rate of 85.17%. As already mentioned before, XLNet allows predicting deep contexts effectively; therefore, their combination allows predicting information correctly.

The accuracy of ERNIE was noted to be at 85%, which was slightly better than that of ELECTRA. This was due to the better ability of ERNIE to understand the context. The ability of ERNIE to understand the context is better because it uses several inputs in the process of training. This enables ERNIE to better understand the context of disease and symptoms. Therefore, ERNIE can make predictions in complex cases of diseases.

Category	Model	Precision	Recall	F1-Score	Accuracy
LLM	XLM- Roberta	0.84	0.83	0.83	0.8331
	ERNIE	0.85	0.84	0.83	0.8392
	ELECTRA	0.87	0.85	0.85	0.8542
DL	Multilayer Perceptron (MLP)	0.87	0.86	0.86	0.8671
	TabNet	0.86	0.85	0.86	0.8427
	Residual MLP	0.87	0.85	0.86	0.8543

When comparing all the three models, it can observe that the ERNIE and XLM-RoBERTa slightly outperform ELECTRA in terms of the accuracy. However, its difference is not very large, which means all the three models are capable of a handling the disease prediction task effectively. This also indicates that dataset and preprocess steps were well-designed, allowing the different models to achieve similar performance levels.

Overall, the results suggest that the while advanced models like ERNIE and XLM-RoBERTa provide slightly better accuracy, ELECTRA still the remains a strong and efficient model. All three models show good generalization ability and can be the effectively used in the symptom-based disease prediction system.

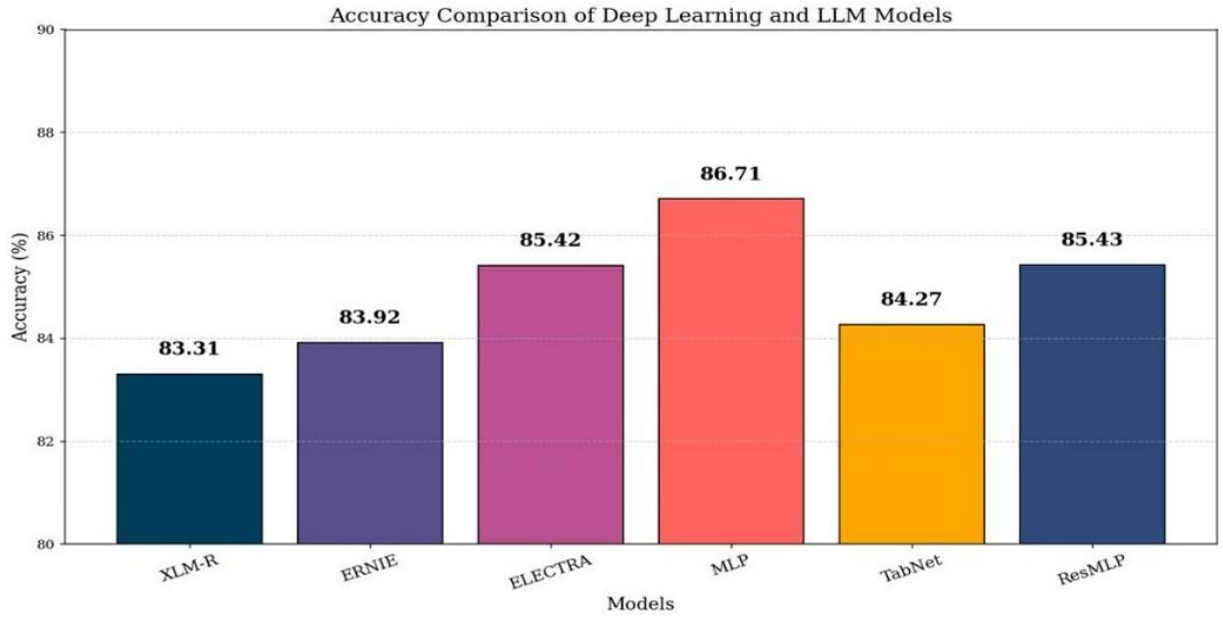


Fig1 - 4.2 Result Analysis for all Models for DL and LLM

Figure: Overall Accuracy Comparison for the Models Used in the Project

From the above figure, it is evident that all the models are performing equally well since the accuracy obtained ranges from 83% to 87%. Thus, the system can accurately predict the diseases using different model techniques.

To begin with, the models under Large Language Models (LLM), XLM-RoBERTa performed the least with an accuracy value of around 83.31%, making it the worst-performing model among all the others. Nevertheless, it performs consistently with reliable results, as it effectively understands the disease from the symptom-based input. On the other hand, ERNIE, which is a relatively advanced model from BERT and XLM-RoBERTa, gave higher accuracy, which is 83.92%, thereby having the capability of better analyzing the relationship between symptoms and diseases. The most effective model of all LLMs was ELECTRA with an accuracy of around 85.42%.

Regarding Deep Learning models, it can be seen that Deep Learning models have better performance than the LLM-based models in this particular project. The accuracy level obtained by

MLP has been observed to be at around 86.71%. Thus, this makes the MLP model the most accurate model in this project.

It signifies that MLP is quite proficient in identifying patterns in structured symptom data. An accuracy of around 84.27% was observed in the TabNet model, which means that it is relatively less effective compared to other deep learning models. However, an accuracy of around 85.43% was achieved in the Residual MLP model, which is quite close to the ELECTRA model.

From the above comparison of accuracy levels obtained from different models, it can be concluded that Deep Learning models especially MLP and Residual MLP perform relatively better than LLM-based models in this particular project. One reason could be that structured data has been used in this particular project, and deep learning models work well with such types of data. However, LLM models are preferred when dealing with unstructured data.

A note worth point here is that the gap between accuracies of these models is relatively small. Most of models are within narrow margin, indicating that all of the models have the potential to solve the task successfully. It further implies that the data preprocessing stage and the dataset itself hold great significance in obtaining such results.

In conclusion, from the graph above, we can observe that all of the models do a fairly decent job, but MLP is the best performing one, followed by Residual MLP and ELECTRA.

4.3 Observations from the work

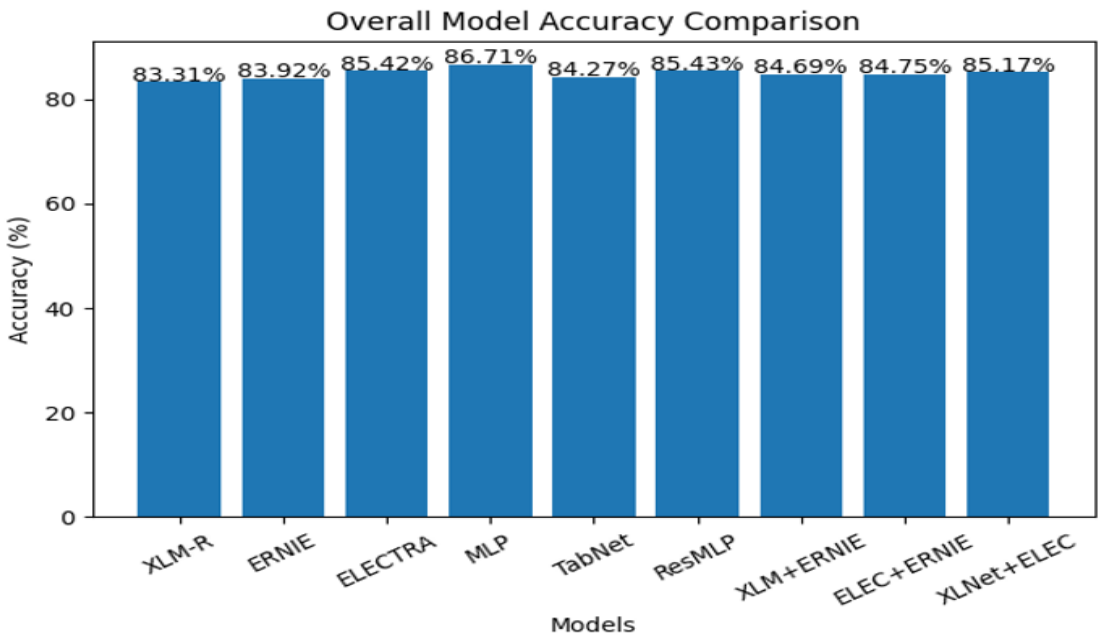


Fig1 - 4.3 Comparision of results for all Models

The figure shows to the represents the all-accuracy comparison of all models uses in the project, including LLM, DL models, and Hybrid models. From graph, it can be clearly observed the all models perform with in close accuracy range, rough between range of 83% and 87%. This can be indicates that the system is the well-designed and all models are capable of the learning meaningful patterns from dataset.

Among the models, the Multilayer Perceptron (MLP) achieve highest accuracy of around 86.7%, making it the best-performing model in this study. This shows that MLP is highly effective in handling structured symptom data & can capture relationships between symptoms and a disease efficiently. Residual MLP also the performs well with an accuracy of around 85.5%, showing that the add residual connections helps improve the model stability.

When looking at the LLM-based models, ELECTRA performs the best with an accuracy of about 85.42%, followed by ERNIE and XLM-RoBERTa. Although LLM models are designed for handling

text data, their performance is slightly lower compared to Deep Learning models in this case. This may be of because the dataset is structured and the better suited for deep learning approaches.

Hybrid models like XLNet + ELECTRA, ELECTRA + ERNIE, & XLM-RoBERTa + ERNIE show competitive performance. Among them, the XLNet + ELECTRA achieves highest accuracy of around 85.17%. These models combine the strengths of multiple architectures, which helps in improving prediction capability, but the improvement is relatively small.

Another important observation from the graph is that the difference in accuracy among the models is the not very large. Most models fall the within a narrow range, which indicates that the dataset quality and preprocessing steps are strong.

Overall, the graph clearly shows the while all models of the perform well, DL models especially MLP provide best results in this project. It also highlights that hybrid models offer slight improvements, and LLM model that can be still remain effective for the text-based of prediction tasks.

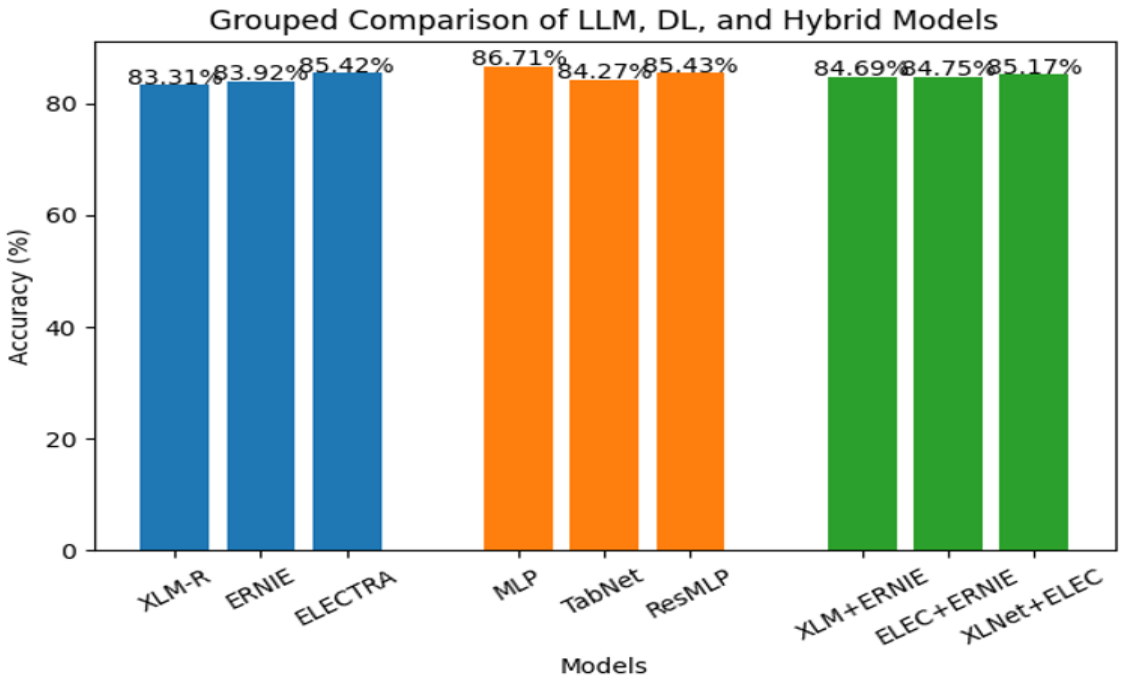


Fig2 - 4.3 Comparision of results for all Models

They is grouped bar chart provides the clearer comparison by the dividing the models into the three categories: Large Language Models, Deep Learning models, and Hybrid LLMs. This group makes it easier to the understand how each category performs the relative to the others.

In the first group represents LLMs, ELECTRA shows highest accuracy among the three, followed by the ERNIE, XLM-RoBERTa. All LLMs perform reasonably well, the accuracy value is slightly lower compared to other categories. The suggests that while LLMs is strong in handling the natural language input, they may the best choice for the structured datasets.

The second group will the represents DL models, where MLP clearly stands out with the highest accuracy of around 87%. Residual MLP also shows strong performance, while TabNet performs slightly lower but the still maintains the good results. This group achieves the highest accuracy among all categories, indicating that the DL models are more suitable for the type of data.

The third group which the represents Hybrid models, and combine the two different architectures. XLNet + ELECTRA performs best, followed by the ELECTRA + ERNIE, XLM-RoBERTa + ERNIE. These are the models show slightly better performance than that some individual LLM models, which means combining of the models can be help improve results. However, the improvement is not of the very significant when compared to the Deep Learning models.

Another important observation from this graph is that the performance across all groups is very quite consistent. There are no extremely variations, which indicates the system is stable and all the models are capable of the handling prediction task is effectively.

This grouped combatively clear shows the Deep Learning models performs best, followed by using the Hybrid models, and then the LLM models. This helps in the selecting the most suitable model for system and provides a clear understanding of how the different approaches compare of with each other.

Chapter – 5

Conclusion and Future work

5.1 Conclusion

In this project, the symptoms-based disease prediction system was successful designed and implement the using different artificial intelligence. The main objective of work was built a system that can take the symptoms as input from users & provide accurate disease predictions in simple and efficient way. The project mainly focused on combining different approaches such as Deep Learning models, Large Language Models to analyses their performance and identify the most suitable method.

The development of system started with the collecting of the large dataset that contains information most about symptoms and corresponding diseases. The dataset includes a wide range of the diseases, which helps in making system more reliable and scareble. Several preprocessing steps are applied to the improve quality of data. This can be included cleaning the dataset, handles the missing values, converting the data into suitable formats for different models. In the LLM-based models, the symptom data was converted into the text data, while for Deep Learning models, it was the maintained in the structured format. This preprocessing step played important role in improving the performance of models.

Different models were implemented and tested in part of system. LLM-based models such as XLM-RoBERTa, ERNIE, and ELECTRA are used to understand and the process symptom descriptions in natural language. These models are known for their ability to capture contextual meaning from text, which makes them suitable for the chatbot-based systems. At same time, Deep Learning models such as Multilayer Perceptron (MLP), TabNet, and Residual MLP were used to work with the structured data. These models are effective in the learning patterns from numerical inputs and widely used for classification tasks.

In addition of individual models, hybrid models were developed by the combining different architecture. The main idea is behind using hybrid models was to take of the advantage of the strengths of multiple models and also improve prediction performance. Models like XLNet + ELECTRA, ELECTRA + ERNIE, and XLM-RoBERTa + ERNIE are tested and also the compared with the individual models.

The performance of the models was evaluated using the standard metrics such as accuracy, precision, recall, and F1-score. From results, it was observed that all the models are performed within the close range of, generally between 83% and 87% accuracy. That can be indicates that the system is the stable and capable of the making reliable predictions. Among LLM models, ELECTRA shows the better performance compared to the ERNIE and XLM-RoBERTa, mainly due to its the efficient training mechanism and the strong classification capability.

When we are comparing Deep Learning models, MLP achieved highest accuracy of around 86.71%, to making it best-performing model in the project. Residual MLP also showed the strong performance, while the TabNet performed slightly lower but still provided the good results. This shows that are Deep Learning models are highly effective for structured datasets and can a capture relationship between symptoms and diseases efficiently.

The hybrid models are performed very well, with accuracy values close 85%. Among them, XLNet + ELECTRA achieved best performance. These models showed the combining different architecture can be slightly improve results by capturing multiple aspects of the data. However, the improvement was not very significant compared to the individual models, which suggests a well-trained single model can also the provide strong performance.

Another the important observation from results is the models showed stable training behaviours. The accuracy increased gradually and the stabilized, while the loss decreased and remained to low. The difference between the training and testing results was small, which indicates that the models are not overfitting and are able to generalize well to new data. This is an important factor in real-world applications, as system needs to perform well on unseen inputs.

The system is also demonstrated the ability to the handle different type of symptom inputs provided consistent predictions. This shows the overall design of the system, including data preprocessing, model selection, and evaluation, are effective. The use of the explainability techniques like SHAP further improve system by the providing insights into how the predictions are made to making system transparent.

The results obtained from precision, recall, and F1-score measurements confirm that all the models perform better and make balanced and reliable predictions. The proposed techniques can be combined with XAI techniques such as SHAP, which is highly efficient for analysis of the decision-making process for the disease prediction using proposed models.

This is highly efficient for the decision-making process, especially for healthcare applications. It can clear the proposed method is highly efficient to the combine efficient Deep Learning algorithms and XAI algorithms to obtain better results for disease prediction and make of appropriate decisions to the develop more accurate & efficient disease prediction systems.

From analysis, it can conclude that Deep Learning models, especially MLP, provide the best performance for the problem. LLM-based model also performs well and are useful for handling natural language input, while the hybrid models offer slight improvements by combining multiple approaches. Each type of the model has own advantages, and choice of model depends in nature of the data and requirements of system.

Over all the summary of this project successful to achieved its objective of building an effective disease prediction system. The results show that the system is capable of providing accurate, the reliable predictions, making them to useful for basic health care support and also early disease detection. The combination of different models and techniques provides a strong foundation for developing intelligent healthcare applications.

5.2 Future study

Even though proposed system shows the good performances to predicting diseases based on symptoms, there is still scope for the further improvement. Future work can focus making the system have the accurately, scalable, and suitable for real-world healthcare applications.

One important area for future improvement to use of larger and more diverse datasets. In this project, the dataset covers a wide range of diseases, but in real-world scenarios, medical data can be much more complex. Including more real clinical data, patient history, and detailed symptom descriptions can help improved accuracy of the models. Integrating datasets from the different sources such as hospitals, health organizations can make the system more robust.

Another possible extension is improving the handling of natural language input. Currently, the system can be processing symptom descriptions, but users may provide input in different formats, languages, incomplete sentences. Future work can be focused making the system more flexible by supporting multilingual input and better understanding of informal or conversational language. This can be achieved using more advanced LLMs/fine-tuning models on domain-specific medical data.

Performance of hybrid models can be further explored. In this project, hybrid models showed slight improvement, but there are still potential to design more effective combinations. Future work can be focused on developing better architectures that combine the strengths of different model more efficiently. Techniques such ensemble learning, attention-based model fusion can explored improve prediction performance.

Another important aspect is improving explain of system. Although SHAP was used in the project, there is still scope to enhance how explanations are presented to users. Future systems can provide more user-friendly explanations, such as highlight the important symptoms or giving reasoning behind predictions. This may increase user trust and make system more useful technical user.

The system can be extended by integrating additional feature such as severity analysis and recommendation systems. For an example, based on the predicted disease, the system can suggest basic precautions, lifestyle changes. This would make the system more practical and helpful for users.

Another area of improvement is real-time deployment. In future, the system can be deployed as a web application/mobile application, allowing users can access it easily. Integration with cloud platforms can help in handling large-scale data & providing faster responses. This would make the system more accessible and for suitable for real-world usage.

There is also scope to improve the efficiency of the models. Some models, especially LLM-based ones, require more computational resources. Future work mainly focus on optimizing these models can reduce memory usage improve speed, making more suitable for deployment in low-resource environments.

Finally, future work can focus on ensuring better reliability and safety of the system. Since healthcare applications require high accuracy, additional validation techniques, testing on real-world data can be carried out. This will help improving the reliability of predictions & making system more dependable.

Overall, the future study focuses on improve data quality, model performance, and deployment aspects. By addressing these areas, the systems can be further developed into a more advance and practical healthcare supporting tool.

References

1. Biswas, A., Pal, P., Bhattacharjee, S., Debnath, S., Chowdhury, S.N., Saha, R.: An explainable AI framework for Parkinson's disease prediction using XGBoost, SHAP, and large language model integration. In: Proceedings of the 12th Interna- tional Conference on Next Generation Computing, Communication, Systems and Security, pp. 8–15 (2025)
2. Banerjee, S., et al.: Large language modeling and classical AI methods for the future of healthcare. Journal of Medicine, Surgery, and Public Health 1, 100026 (2023)
3. Tabassum, M., Kaiser, M.S.:Towards explainable conversational AI for early diag- nosis with large language models.arXiv preprint arXiv:2512.17559 (2025)
4. Zagade, A., et al.: AI-based medical chatbot for disease prediction (2024)
5. Yu, H., McGuinness, S.: An experimental study of integrating fine-tuned LLMs and prompts for enhancing mental health support chatbot system. Journal of Medical Artificial Intelligence, 1–16 (2024)
6. Isijola, A.O., et al.: Medical Diagnosis and Treatment Recommendations (Doc-Bot) using Large Language Models. FUOYE Journal of Engineering and Technology 9(4), 609–614 (2024)
7. Chakraborty, S., et al.: An AI-based medical chatbot model for infectious disease prediction. IEEE Access 10, 128469–128483 (2022)
8. Sivasamy, S.: AI-Driven Medical Chatbot for Predicting and Managing Infectious Diseases. International Journal of Advanced Research in Science, Communication and Technology, 772–777 (2025)
9. Biradar, S., Shastri, S.: Medical Chatbot: AI Based Infectious Disease Prediction Model. Journal of Scientific Research and Technology, 1–12 (2024)
10. Barath Lakshman, A.N.: DocBot: Integrating LLMs for Medical Assistance (2024)
11. Zhang, S., Song, J.: A chatbot based question and answer system for the auxiliary diagnosis of chronic diseases based on large language model. Scientific Reports 14(1), 17118 (2024)
12. Yang, X., et al.: Application of large language models in disease diagnosis and treatment. Chinese Medical Journal 138(2), 130–142 (2025)
13. Badempet, P., Cheerala, P., Anagondi, S.P.: A healthcare system using machine learning techniques for disease prediction with chatbot assistance. ScienceOpen Preprints (2023)

14. Zheng, H., et al.: Medication recommendation system based on natural language processing for patient emotion analysis. Academic Journal of Science and Technol- ogy 10(1), 62–68 (2024)
15. Ahmed, S.T., et al.: Medical ChatBot assistance for primary clinical guidance using machine learning techniques. Procedia Computer Science 233, 279–287 (2024)
16. Kevin, B., Vikin, B., Nair, M.: Building a Chatbot for healthcare using NLP. Authorea Preprints (2023)
17. Kim, Y., et al.: Predicting medical specialty from text based on a domain-specific pre-trained BERT. International Journal of Medical Informatics 170, 104956 (2023)
18. Yang, R., et al.: Large language models in health care: Development, applications, and challenges. Health Care Science 2(4), 255–263 (2023)
19. Ma, R., et al.: Multimodal machine learning enables AI chatbot to diagnose oph- thalmic diseases and provide high-quality medical responses. npj Digital Medicine 8(1), 64 (2025)12
20. Silvestri, C., et al.: Evaluation of a novel large language model (LLM)-powered chatbot for oral boards scenarios. Global Surgical Education 3(1), 1–12 (2024)
21. Xu, N.: The investigation of symptom to disease chatbot based on LLAMA-2. Applied and Computational Engineering 55, 39–43 (2024)
22. Babu, A., Boddu, S.B.: BERT-based medical chatbot: Enhancing healthcare com- munication through natural language understanding. Exploratory Research in Clinical and Social Pharmacy 13, 100419 (2024)

APPENDIX

```python
model_name = "nghuyong/ernie-2.0-en"
max_length = 128

tokenizer = AutoTokenizer.from_pretrained(model_name)

print("Tokenizing train data...")
train_encodings = tokenizer(
    X_train,
    truncation=True,
    padding=True,
    max_length=max_length,
    return_tensors="pt"
)

print("Tokenizing test data...")
test_encodings = tokenizer(
    X_test,
    truncation=True,
    padding=True,
    max_length=max_length,
    return_tensors="pt"
)
```

```python
base_model = AutoModelForSequenceClassification.from_pretrained(
    model_name,
    num_labels=num_labels
)

base_model = base_model.to(device)

if torch.cuda.is_available() and torch.cuda.device_count() > 1:
    print(f"Using {torch.cuda.device_count()} GPUs with DataParallel")
    model = nn.DataParallel(base_model)
else:
    model = base_model

print("Model device:", next(model.parameters()).device)
```

Fig-6: Code Snippet of LLM model

*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Disclaimer

Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?

The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?

Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

iThenticate Page 2 of 4 - Integrity Overview Submission ID trn:oid:::3618:135428788

18% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

- **254** Not Cited or Quoted 20%
  Matches with neither in-text citation nor quotation marks
- **5** Missing Quotations 0%
  Matches that are still very similar to source material
- **4** Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

14% Internet sources
16% Publications
0% Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.